Reviewed Preprint
v1 • December 17, 2025
Not revised

For correspondence:
Andreas.Mayer@unil.ch

Competing interests: No competing interests declared

Funding: See page 27

Reviewing editor: Francesca Bottanelli, Freie Universität Berlin, Germany

CROP2, a Retriever-PROPPIN Complex Mediating Protein Export from Endosomes to the Plasma Membrane

Maria Giovanna De Leo, Andreas Mayer

Department of Immunobiology, University of Lausanne, Lausanne, Switzerland

eLife Assessment

The authors present evidence for a WIPI2-Retriever complex (termed CROP2) that couples cargo selection to carrier fission at endosomes. CROP2 appears to function analogously to the previously described CROP1 complex, formed by WIPI1 and Retromer, with which it shares structural similarities. They provide **convincing** evidence that CROP1 and CROP2 regulate the trafficking of distinct subsets of cargoes; however, the cellular evidence for the existence of these distinct complexes remains **incomplete**. Overall, the findings are **important** and expand our understanding of how cargo selection by Retriever and Retromer is orchestrated at endosomes.

https://doi.org/10.7554/eLife.109403.1.sa3

Abstract

Endosomes generate tubulo-vesicular carriers to redistribute proteins between plasma membrane, Golgi, and lysosomes. These transport routes employ distinct combinations of sorting nexins with complexes such as Retromer or Retriever. We now show that, while Retromer associates with the PROPPIN WIPI1 to form the previously described CROP complex, Retriever associates with WIPI2, forming CROP2. WIPI2 integrates into Retriever-dependent coat complexes, since it interacts both with the Commander subunit CCDC93 and its cognate sorting nexin SNX17. CROP and CROP2 are exclusive in their physical associations and pathway selective. Whereas CROP2 is required for endosomal exit of β1-Integrin, it does not affect CROP-dependent cargos, such as EGFR or GLUT1. Vice versa, CROP is not required for β1-Integrin trafficking. WIPI1 and WIPI2 rely on similar molecular features. They use the same FSSS motif to bind to Retromer and Retriever, respectively, and an amphipathic membrane-inserting α-helix, which conveys membrane fission activity to PROPPINs. This suggests that Retromer and Retriever coats integrate distinct PROPPIN isoforms to promote fission of the respective endosomal carriers formed by them.

Introduction

Endosomal membrane traffic regulates the abundance and activity of many receptors and transporters at the plasma membrane, which renders it critical for the development and homeostasis of cells and tissues (McNally and Cullen, 2018). In line with this, perturbations in endosomal trafficking contribute to a wide range of diseases, including cancer and some of the most common neurological disorders (Carosi et al., 2023; Mellman and Yarden, 2013; Schreij et al., 2016). Many plasma membrane proteins cycle between the cell envelope, endosomes, and the Golgi. They can move on to lysosomes for degradation unless they are retrieved before endosomes fuse with lysosomes (Cullen and Steinberg, 2018). Endosomes thus serve as sorting stations at an intersection between several compartments. To collect cargo proteins, endosomes employ a variety of coats that form tubulo-vesicular carriers. Many of them are based on sorting nexins, which scaffold the membrane into narrow tubules. Sorting nexins can act on their own, such as in the ESCPE-1 complex (Kvainickas et al., 2017; Lopez-Robles et al., 2023; Simonetti et al., 2019, 2017), or together with complexes interconnecting them, such as Retromer or Retriever and the Commander complex, into which Retriever can be integrated (Bartuzi et al., 2016; Boesch et al., 2023a; Chandra et al., 2024; Healy et al., 2023; Mallam and Marcotte, 2017; McNally et al., 2017; Phillips-Krawczak et al., 2015; Seaman, 2021). Further coats, such as clathrin adapter complexes, VINE, and ACAP1-clathrin can also act on endosomes (Dai et al., 2004; Hooy et al., 2022; Li et al., 2007; Pang et al., 2014; Ren et al., 2013; Sachse et al., 2002; Shortill et al., 2022). These coat components organize a variety of endosomal protein exit routes with partially overlapping cargo specificity (Bean et al., 2017; Kvainickas et al., 2017; McGough et al., 2014; Steinberg et al., 2013).

Retromer is part of a protein coat mediating protein exit from endosomes in association with sorting nexins, such as Vps5, Vps17 or Snx3/Grd19 in yeast, and SNX27, ESCPE-1 or SNX3 in human cells (Seaman, 2021). Retromer consists of a VPS26– VPS29–VPS35 heterotrimer, which can form higher order complexes through VPS35 and VPS26 dimerization (Collins et al., 2008; Deatherage et al., 2020; Hierro et al., 2007; Kendall et al., 2022, 2020; Kovtun et al., 2018; Leneva et al., 2021; Lucas et al., 2016; Purushothaman et al., 2017). While sorting nexins themselves change the spontaneous curvature of membranes and can tubulate them (Kovtun et al., 2018; Leneva et al., 2021; van Weering et al., 2012; Zhang et al., 2021), this activity can be enhanced by the association with Retromer, which may provide additional driving force for scaffolding the membrane (Gopaldass et al., 2023).

In human cells, numerous cargos recycle back to the cell surface in a Retromer- independent manner (Steinberg et al., 2013). Many of these depend on Retriever, an evolutionarily conserved complex of VPS26C/DSCR3, VPS29 and VPS35L/C16orf62, proteins that show distant homology to the Retromer subunits (McNally et al., 2017). Whereas the overall organization and shape of Retriever resemble those of Retromer, Retriever shows different cargo specificity and distinct features. Retriever forms part of a much larger complex, Commander, which integrates Retriever with the 12-subunit CCC complex and DENND10 and cooperates with the sorting nexin SNX17 to form coats (Boesch et al., 2023a; Butkovič et al., 2024; Healy et al., 2023; Martín-González et al., 2024; Singla et al., 2024; Steinberg et al., 2012). Like Retromer, also Commander can interact with WASH and thereby link to actin (Derivery et al., 2009; Gomez and Billadeau, 2009; Guo et al., 2024; Harbour et al., 2012; Jia et al., 2012; Phillips-Krawczak et al., 2015; Seaman et al., 2009).

Cargo carriers detach from endosomes by membrane fission (Gopaldass et al., 2024; Naslavsky and Caplan, 2023). Several factors have been implicated in this detachment, among them actin and the actin-controlling WASH complex (Derivery et al., 2009; Frisby et al., 2024; Gomez and Billadeau, 2009; Harbour et al., 2012; Jia et al., 2012; Seaman et al., 2009), dynamin-like GTPases and ATPases such as Vps1 and EHD1 (Arlt et al., 2015; Chi et al., 2014; Daumke et al., 2007; Deo et al., 2018; Gokool et al., 2007; Grant et al., 2001; Solinger et al., 2020), the ESCRT-III subunit IST1 (Clippinger et al., 2024) and the PROPPIN (β-propeller that binds phosphoinositides) WIPI1 and its yeast homolog Atg18 (DeLeo et al., 2021; Gopaldass et al., 2024, 2017). Coats and actin can exert forces on the underlying lipid bilayer, which not only allow the tubule to grow but may also produce friction between lipids and protein coats that contributes to fission (Johannes et al., 2014; Simunovic et al., 2018, 2017). In addition, dynamin-like proteins actively deform the bilayer to compress and destabilize it (Antonny et al., 2016). The lipid phosphatidylinositol-(3,5)- bisphosphate (PI(3,5)P2) and/or PI5P is also required for fission (DeLeo et al., 2021; Giridharan et al., 2022; Gopaldass et al., 2017; Rivero-Ríos and Weisman, 2022; Zieger and Mayer, 2012). PROPPINs are bona-fide effectors of PI(3,5)P2 (Courtellemont et al., 2022; DeLeo et al., 2021). They associate with membranes through two phosphoinositide binding sites (Baskaran et al., 2012; Krick et al., 2012; Liang et al., 2019; Proikas-Cezanne et al., 2004; Vicinanza et al., 2015; Watanabe et al., 2012). Upon membrane binding, a hydrophobic loop, which is located between the two lipid binding sites, folds into an amphipathic α-helix that inserts into the bilayer. Thereby, PROPPINs can tubulate lipid bilayers and drive their fission (Courtellemont et al., 2022; DeLeo et al., 2021; Gopaldass et al., 2017; Gopaldass and Mayer, 2024).

PROPPINs are a protein family that binds phosphatidylinositol 3-phosphate (PI3P), phosphatidylinositol 5-phosphate (PI5P) and phosphatidylinositol 3,5-bisphosphate (PI(3,5)P2) (Baskaran et al., 2012; Busse et al., 2015; Dove et al., 2004; Gopaldass and Mayer, 2024; Jeffries et al., 2004; Vicinanza et al., 2015). They are conserved from yeast (Atg18, Atg21, and Hsv2) to humans (WIPI1, WIPI2, WIPI3/WDR45B and WIPI4/WDR45) and exist in most eukaryotes in several isoforms, which can be grouped into two evolutionary clades (Dove et al., 2004; Proikas-Cezanne et al., 2015). In mammals, Atg18, Atg21, WIPI1 and WIPI2 belong to one clade, Hsv2, WIPI3 and WIPI4 to the second (Cong et al., 2021; Polson et al., 2010). WIPI1 and WIPI2 concentrate on phagophores when autophagy is induced and they are necessary for efficient formation of autophagosomes (Dooley et al., 2014; Polson et al., 2010; Proikas-Cezanne et al., 2004). WIPI2, which has a stronger impact on autophagy, interacts with different key factors for autophagosome formation.

WIPI1 and Atg18 also support membrane fission and protein sorting on endosomes and lysosomes, but for these functions they rely on molecular features that are distinct from those important to autophagy (Courtellemont et al., 2022; DeLeo et al., 2021; Gopaldass et al., 2017; Gopaldass and Mayer, 2024). WIPI1 localizes to endo- lysosomal compartments, where it affects the size of endosomes and the exit of proteins from endosomes to lysosomes (EGF-receptor), to the Golgi (Shiga-toxin), or to the cell surface (Transferrin receptor, GLUT-1) (DeLeo et al., 2021; Jeffries et al., 2004). Based on the differential effects of substitutions in the two phosphoinositide binding sites of WIPI1 it was suggested that binding of PI3P allows WIPI1 to support the formation of endosomal transport carriers while PI(3,5)P2 and lipid binding site two are needed for membrane fission at endosomes (DeLeo et al., 2021). WIPI1 as well as its yeast homolog Atg18 associate with Retromer to form the CROP complex, which strongly augments the fission activity of the PROPPIN and makes its binding to membrane PI(3,5)P2 dependent (Courtellemont et al., 2022; Marquardt et al., 2023).

We have recently identified the Retromer- and WIPI1-containing CROP complex as an agent supporting the export of multiple cargos from endosomes, comprising both Retromer-dependent and Retromer-independent exit routes (Courtellemont et al., 2022; DeLeo et al., 2021). Since not all cargo exit from endosomes depends on WIPI1 and CROP we analysed alternative PROPPINs that might be involved in WIPI1- independent exit from endosomes. This led us to uncover a novel role for a WIPI2- Retriever complex in trafficking of β1-Integrin, which we present below.

Results

WIPI1 promotes the exit of diverse cargos from endosomes, for example for transporting Transferrin to the plasma membrane, GLUT1 to the plasma membrane, EGF-receptor (EGFR) to lysosomes, and of Shiga toxin to the Golgi (DeLeo et al., 2021). β1-Integrin provides an exception, its delivery to the cell surface being unaffected by knockout of WIPI1 (Courtellemont et al., 2022). We investigated whether the endosomal exit of β1-Integrin might depend on another PROPPIN. Since WIPI1 and WIPI2 are similar (Bakula et al., 2013; Gopaldass and Mayer, 2024), belong to the same evolutionary clade, share two predicted lipid binding sites as well as the potential to form an amphipathic helix on their CD loop 6 (Cong et al., 2021; Gopaldass et al., 2017; Polson et al., 2010), we focused our attention on WIPI2.

WIPI1 and WIPI2 are specific for distinct trafficking pathways

We tested the impact of WIPI2 knockdown (WIPI2^{KD}) on the afore-mentioned trafficking pathways. siRNA against WIPI2 reduced the protein in whole cell extracts by 94% when compared to cells transfected with control siRNA (WIPI2^{CT}) (Fig. S1). We analysed trafficking of EGFR from early endosomes to degradative, LAMP1- containing compartments. After 24h of serum starvation, cells were stimulated with EGF and analysed for EGFR localization by immunofluorescence. 15 min after EGF addition, EGFR-positive structures colocalized with the early endosomal marker EEA1 to more than 75% in both WIPI2^{KD} and WIPI2^{WT} cells, indicating that EGF had induced efficient internalization of EGFR in both cases (Figure 1). After 30 and 60 min of EGF stimulation, EGFR increasingly shifted from EEA1- towards LAMP1-positive lysosomal structures.

EGFR signals declined after 60 min in the immunofluorescence images as well as in Western blot analyses of the cells (Fig. 1 C-E). Thus, WIPI2 knockdown does not impair EGFR trafficking from EEA1 endosomes to LAMP1-positive compartments and leaves lysosomal degradation of EGFR functional.

To test the role of WIPI2 in Retromer-dependent trafficking we assayed the localisation of the glucose importer GLUT-1 because this protein accumulates on endosomes if its recycling to the plasma membrane through Retromer is impaired (Steinberg et al., 2013). To quantify the surface fluorescence, we acquired z-stacks of non- permeabilized, immunostained cells, generated maximum intensity projections and integrated the fluorescence signal over the entire cell. Non-permeabilized WIPI2^{KD} cells showed similar cell surface staining of GLUT-1 as WIPI2^{CT} cells (Figure 2 A-B), suggesting that also Retromer-mediated recycling of GLUT1 occurs independently of WIPI2.

Next, we investigated the requirement of WIPI2 in the trafficking of β1-Integrin. Surface accumulation of this Integrin subunit is unperturbed by loss of SNX27 or Retromer but depends on the SNX17-Retriever pathway (McNally et al., 2017; Steinberg et al., 2013). Knockdown of WIPI2 induced a loss of β1-Integrin immunofluorescence from the surface of HK2 cells. The effect was comparable to that of a knockdown of SNX17 (Figure 3 A, B). When the plasma membrane of the cells was detergent-permeabilized to improve accessibility of internal structures for immunostaining, cells depleted of WIPI2 showed an accumulation of β1-Integrin on intracellular structures. This signal colocalized with the early endosome marker EEA1 and, to a lesser degree, with the late endosomal/lysosomal marker LAMP1 (Figure 3 C-D).

In accord with previous observations (DeLeo et al., 2021), β1-Integrin surface labelling in non-permeabilized cells was not affected by knockdown of WIPI1 (Figure S2). Thus, β1-Integrin recycling to the plasma membrane selectively requires WIPI2.

The conserved amphipathic α-helical and FSSS motifs of WIPI2 promote β1- Integrin recycling

WIPI2 shares two critical features with other PROPPINs (Gopaldass and Mayer, 2024). These include an FSSS motif in the membrane-distal part and the CD loop on blade 6 of the β-propeller structure. The CD loop promotes membrane fission activity of Atg18 during the fragmentation of yeast vacuoles (Gopaldass et al., 2017) and in the detachment of endosomal carriers (DeLeo et al., 2021). Upon contact with the bilayer, the loop folds into an amphipathic α-helix that is conserved among PROPPINs and necessary for membrane fission activity. To test whether the amphipathic nature of this helix also contributes to the function of WIPI2 in β1-Integrin retrieval to the cell surface, we swapped two pairs of amino acids on opposite sides of this α-helix in WIPI2 (Figure 4 A). This manipulation preserves the amino acid composition of the helix but abolishes its amphipathic character. The resulting ^{EGFP}WIPI2^{SLoop} (scrambled loop) variant and the wildtype version ^{EGFP}WIPI2^{WT} were expressed in WIPI2^{CT} and WIPI2^{KD} cells at comparable levels, as shown by Western blot (Figure S3). Whereas permeabilized WIPI2^{KD} cells displayed β1-Integrin concentrated on perinuclear endosomes, expression of ^{EGFP}WIPI2^{WT} rescued this phenotype (Figure 4 B). ^{EGFP}WIPI2^{Sloop} failed to induce such recovery in WIPI2^{KD} cells. It even exerted a dominant-negative effect when expressed in WIPI2^{CT} cells. (Figure 4 C-D). Cells expressing ^{EGFP}WIPI2^{SLoop} also showed conspicuous long tubules, which were more readily detectable in live cell imaging (Figure S4) than in fixed cells (Figure 4).

Alignments of Atg18 and WIPI1/2 orthologs have revealed an FSS/TS motif in blade 2 (Gopaldass et al., 2017; Gopaldass and Mayer, 2024). On the β-propeller structure of these PROPPINS, this motif is located at the membrane-distal side. It is a potential phosphorylation site and required for the interaction of Retromer with WIPI1 and Atg18 (Courtellemont et al., 2022). Since the overall structural arrangement of Retriever resembles that of Retromer (Bartuzi et al., 2016; Boesch et al., 2023a; Healy et al., 2023; McNally et al., 2017), we tested whether WIPI2 requires its FSSS motif to bind Retriever. To this end, we generated WIPI2^{S67E} and WIPI2^{S67A} to substitute the central serine in the FSSS motif and mimic effects of potential phosphorylation at this site.

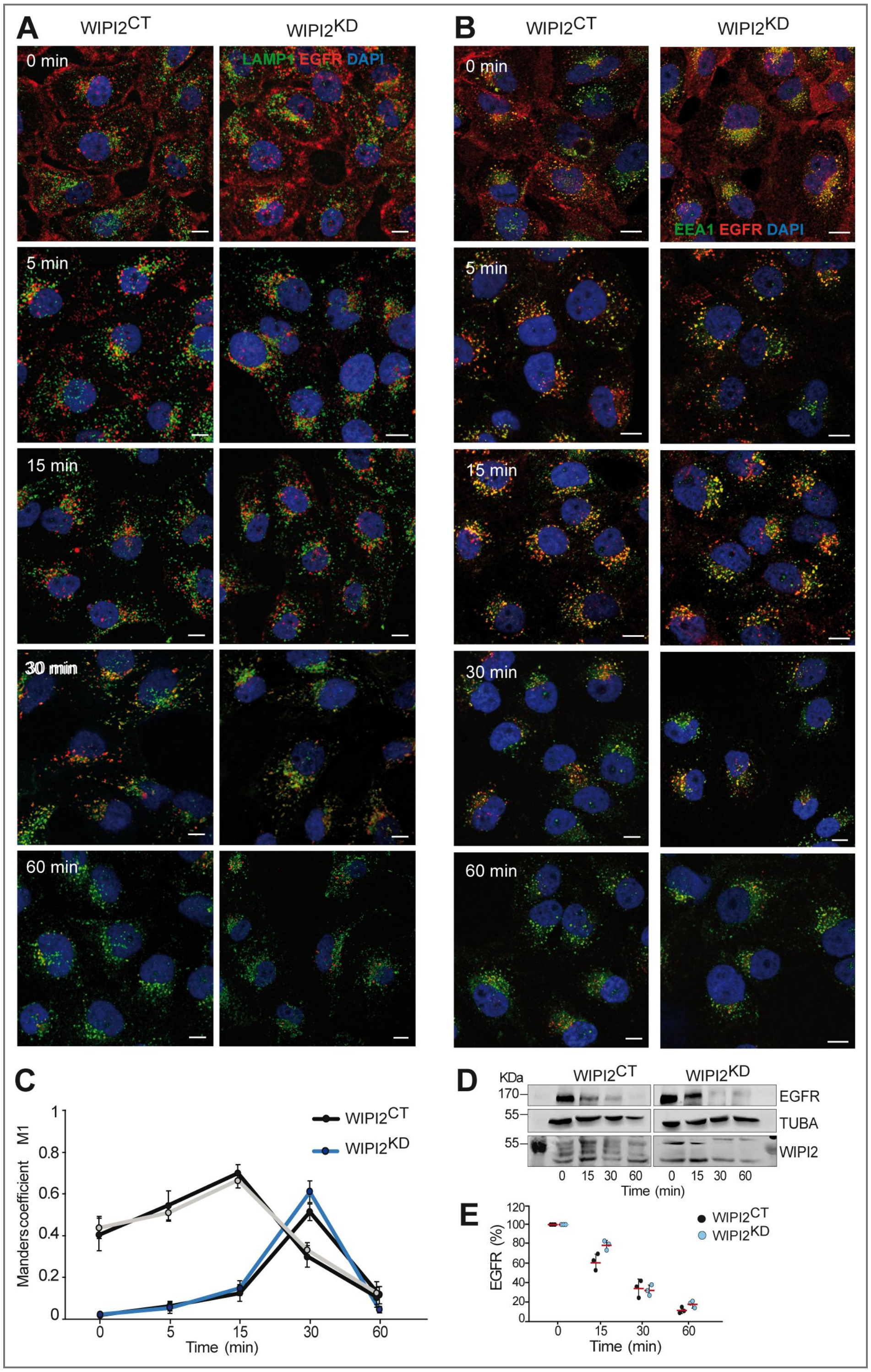

Fig. 1. Effect of WIPI2 knockdown on EGFR degradation.

A, B) WIPI2^{CT} and WIPI2^{KD} cells were serum-starved for 24 h and then supplemented with EGF (100 ng/ml). After the indicated periods of time, cells were fixed, permeabilized, DAPI-labeled (blue) and decorated with antibodies to EGFR (red), EEA1 or LAMP1 (green). Scale bars: 10 µm. **C)** Colocalization of EGFR with EEA1 or LAMP1 was quantified over time using the images from A and B and Manders' correlation coefficients were calculated. Values are the mean ± s.d.; (n = 3 independent experiments). 150 cells were quantified per sample. **D)** EGFR degradation. WIPI2^{CT} and WIPI2^{KD} cells were stimulated with EGF for different periods of time, lysed and subjected to SDS-PAGE and Western blot analysis for EGFR. α-Tubulin served as a loading control. **E**) Quantification of EGFR from (E), using the value at time 0 (cells starved for 24 h) as 100% reference. Data are means ±s. d., from three independent experiments.

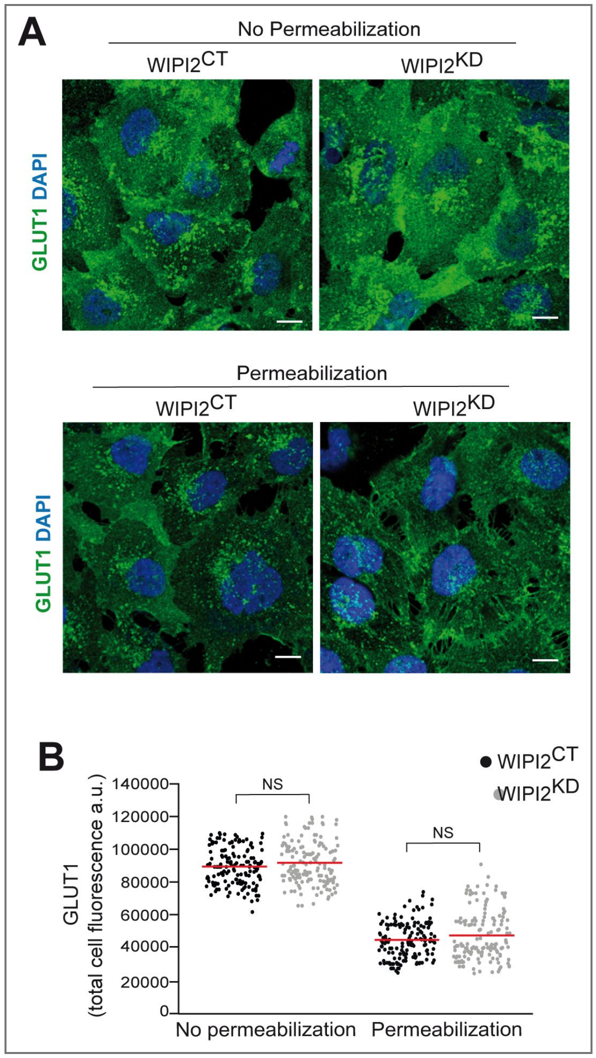

Figure 2. **GLUT1 expression and localization upon WIPI2 knockdown.**

A. GLUT1 cell surface exposure. Control and WIPI2^{KD} cells were fixed and stained with antibody against GLUT1 and with DAPI. Where indicated, cells had been permeabilized with 0.05% before staining. Scale bars: 10 µm. **B.** Quantification of GLUT1-immunofluorescence in cells from A. Regions of interest (ROIs) corresponding to each cell and in some regions outside the cells (background) were manually defined using image J software. Total cell fluorescence was integrated and corrected for background fluorescence. 150 cells per condition, stemming from three independent experiments, were quantified. Red bars show the means. P values were calculated applying an unpaired Student's t-test with unequal variances. The analysis was performed with 99% confidence. NS = not significant ($P > 0.05$).

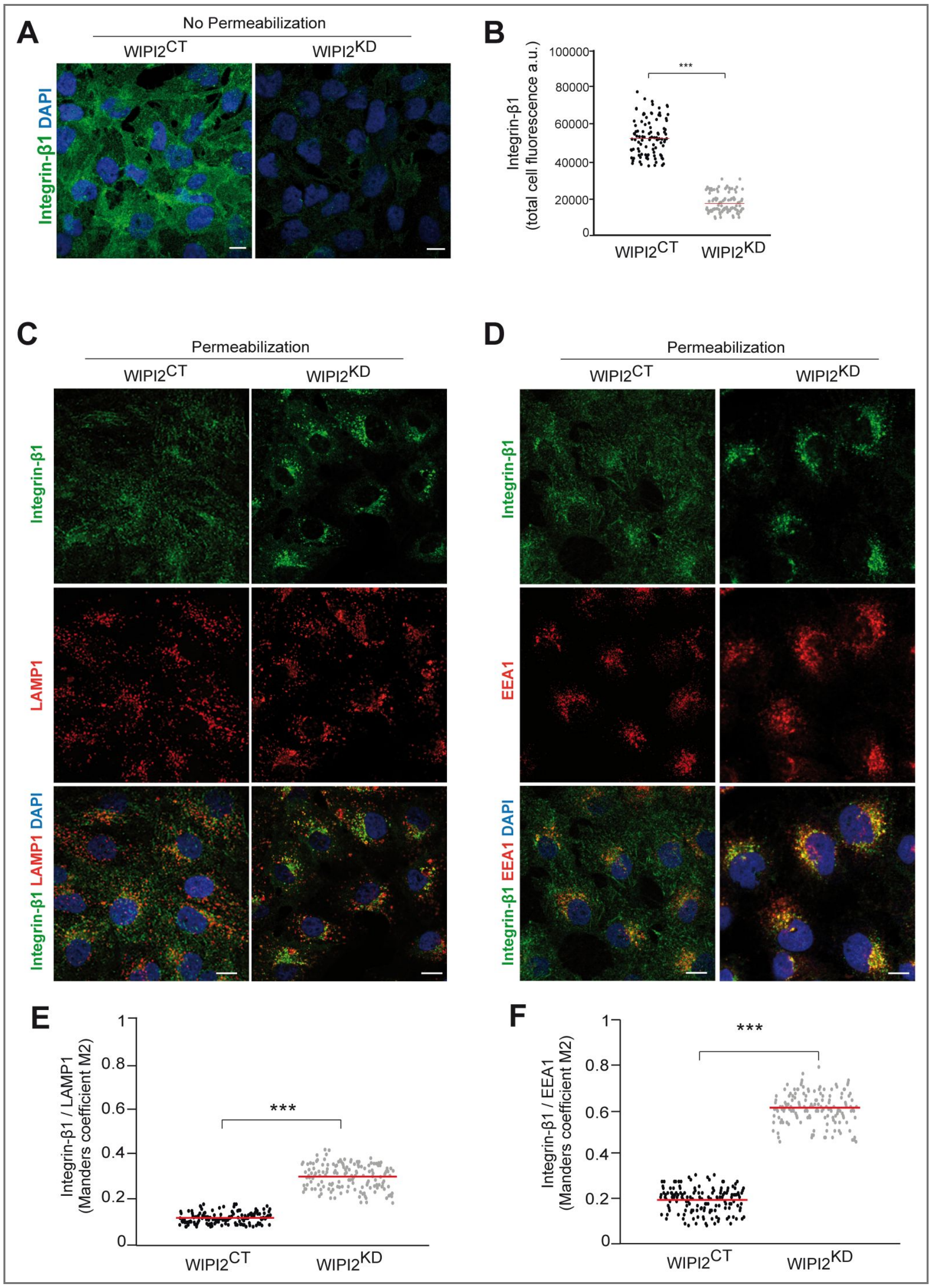

Figure 3. **Impact of WIPI2 knockdown on β1-Integrin localization.**

A. β1-Integrin cell surface expression. Control and WIPI2^{KD} cells were fixed and stained with antibody to β1- Integrin (green) and with DAPI (blue), without detergent permeabilization. Scale bars: 10 mm. B. Quantification of β1-Integrin immunofluorescence in cells from A. Regions of interest (ROIs) corresponding to each cell and in some regions outside the cells (background) were manually defined using ImageJ software. Total cell fluorescence was integrated and corrected for background fluorescence. 150 cells per condition, stemming from three independent experiments, were quantified. Red bars show the means. P values were calculated applying unpaired Student's t-test with unequal variances. The analysis was performed with 99% confidence. ***P < 0.001. C. Immunofluorescence staining of intracellular β1-Integrin (green) and EEA1 (red) or LAMP1 (red) in CTRL and WIPI2^{KD} cells. Cells were fixed, permeabilized with 0.05% saponin and stained. Scale bars: 10 mm. D. Colocalization of β1-Integrin with LAMP1 or EEA1 was measured in cells from C, using Manders' colocalization coefficient M2, calculated in Image J. Colocalization was quantified from three independent experiments with a total of 150 cells. An unpaired Student's t-test with unequal variances was used to calculate P-values. The analysis was performed with 99% confidence. ***P < 0.001.

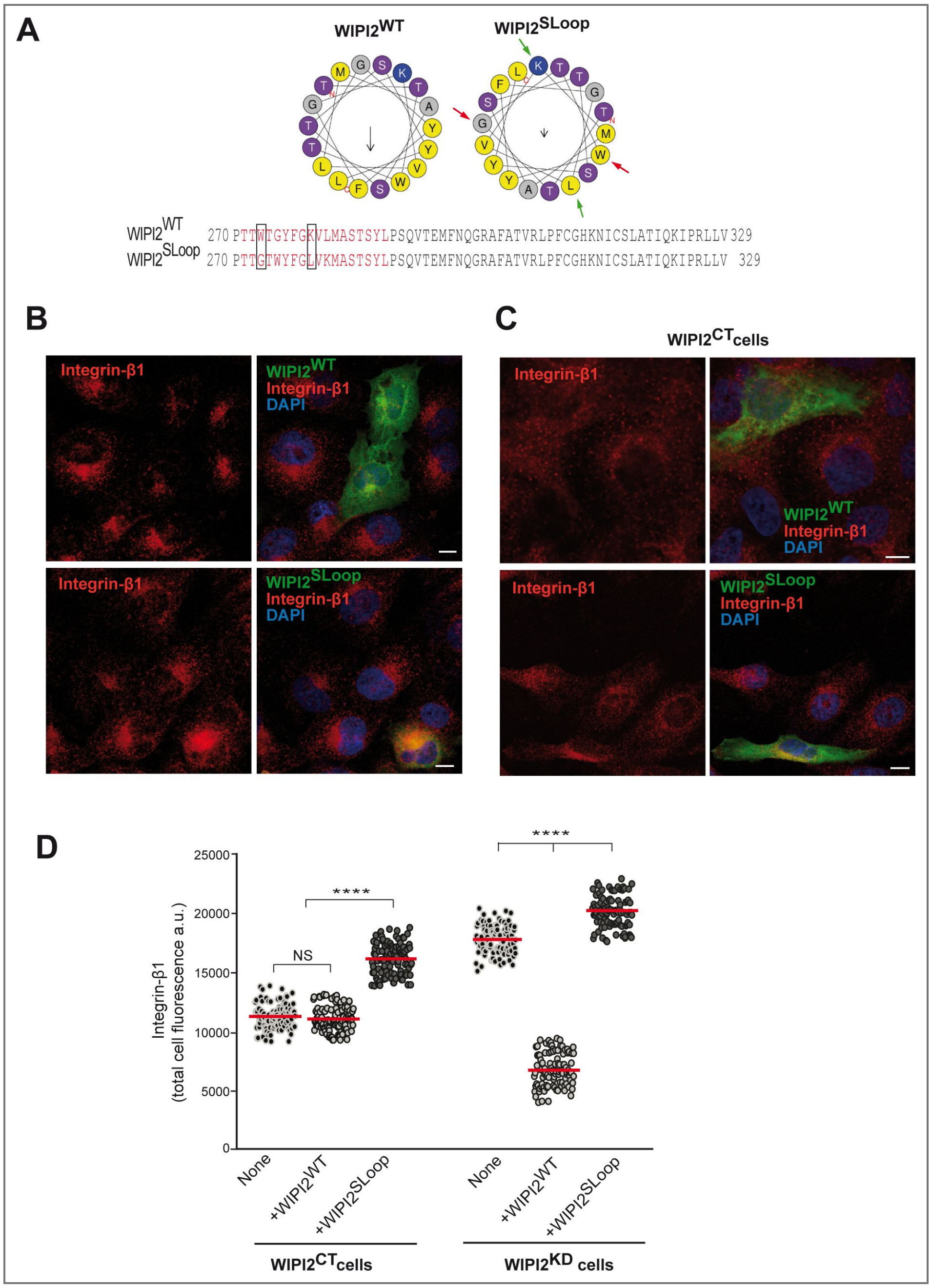

Figure 4. **Role of the amphipathic α-helix of WIPI2 in β1-Integrin sorting.**

A. The amphipathic α-helix. Helical wheel projections showing the CD-loop on blade 6 of the wildtype sequence, $WIPI2^{WT}$, and $WIPI2^{Sloop}$. Coloured arrows indicate the two pairs of amino acids that have been swapped in $WIPI2^{Sloop}$. The magnitude and direction of the hydrophobic moment of the helices was predicted using the online tool Heliquest (Gautier et al., 2008). It is indicated by the vector in the centre of the wheels. Sequences of the hydrophobic loop region of $WIPI2^{WT}$ and $WIPI2^{Sloop}$ are shown, and predicted α-helices are plotted in red. The two pairs of hydrophobic/hydrophilic amino acids that are swapped in $WIPI2^{Sloop}$ are highlighted by rectangles in the sequences. **B,C.** β1-Integrin localisation. $WIPI2^{KD}$ (B) and $WIPI2^{CT}$ (C) cells were transfected with a plasmid carrying siRNA-resistant $^{EGFP}WIPI2^{WT}$ or $^{EGFP}WIPI2^{Sloop}$. After 18h of viral transfection, cells were fixed, permeabilized with 0.05% saponin and stained with DAPI and antibodies to β1-Integrin. Scale bars: 10 mm. **D.** Quantification of β1-Integrin immunofluorescence in cells from B and C. Regions of interest (ROIs) corresponding to cells expressing the indicated WIPI2 variants, and some regions outside the cells (background), were manually defined using Image J software. Total cell fluorescence was integrated and corrected for background fluorescence. 105 cells per condition stemming from three independent experiments were analysed. Red bars indicate the means. P-values were calculated by Welch's t-test. The analysis was performed with 99% confidence: ****$P < 0.00001$. NS: not significant ($P>0.05$)

The functionality of these WIPI2 variants was tested by rescuing $WIPI2^{KD}$ cells through expression of siRNA resistant variants of $^{EGFP}WIPI2$. Both $^{EGFP}WIPI2^{S67E}$ and $^{EGFP}WIPI2^{S67A}$ were expressed to similar levels as an $^{EGFP}WIPI2$ wildtype allele (Figure S3). Immunofluorescence staining of non-permeabilized cells showed that the expression of $^{EGFP}WIPI2^{WT}$ re-established the exposure of β1-Integrin at the cell surface, while $^{EGFP}WIPI2^{S67A}$ and $^{EGFP}WIPI2^{S67E}$ were ineffective (Figure 5 A-C). Expression of the two variants also impaired surface exposure of β1-Integrin in $WIPI2^{CT}$ cells (Figure 5 B-D). This dominant negative effect is significant because it implies that the two variants compete with the endogenous protein. To be able to do so they should be folded, suggesting that the S67 substitutions do not grossly perturb WIPI2 structure. A conspicuous consequence of both of both FSSS substitutions was the accumulation of $^{EGFP}WIPI2^{S67A}$ and $^{EGFP}WIPI2^{S67E}$ in long tubules, which often interconnected several dot-like structures, presumably endosomes (Figs. 5 and S4). This suggests that both alleles might permit the formation of tubulo-vesicular endosomal structures but interfere with their detachment. Taken together, these observations support the notion that WIPI2 contributes fission activity to Retriever- mediated trafficking through its amphipathic helix and its FSSS motif.

WIPI2 interacts with Retriever, Commander and SNX17

Next, we tested whether WIPI1 and WIPI2 physically interacts with Retriever or Retromer and could become part of their respective coats. In that case, WIPI2 should interact not only with Retriever itself, but also with the sorting nexin SNX17 and perhaps the CCC subunits, which integrate Retriever into the Commander complex (Butkovič et al., 2024; Martín-González et al., 2024; Singla et al., 2024). We assayed this by co-immunoadsorption. An HA-tagged allele ($WIPI2^{HA}$) was expressed in HK2 cells using lentiviruses. Extracted proteins were adsorbed to anti-HA beads and analysed by Western blotting (Figure 6). Whereas the Commander subunit CCDC93 and SNX17 co-adsorbed to the beads in extracts from cells expressing $WIPI2^{HA}$ (Figure 6 A), these signals were absent from cells expressing only non-tagged WIPI2. Co-adsorption of SNX17 was abolished by knockdown of the Retriever subunit VPS26C (Figure 6B). Although VPS26C is quite similar to the Retromer subunit VPS26, only VPS26C co-adsorbed with $WIPI2^{HA}$ but VPS26 was not detectable (Figure 6 C-D). The situation was inversed when the HA tag was attached to WIPI1. In this case, VPS26 was recovered on the beads but VPS26C was not (Figure 7A, B). SNX27, a sorting nexin that interacts with Retromer (Simonetti et al., 2022; Temkin et al., 2011), was also co-adsorbed with $WIPI1^{HA}$ (Figure 7C). This suggests that Retriever bridges WIPI2 and SNX17 and that Retromer bridges WIPI1 with SNX27. These PROPPINs might thus be selectively integrated into the respective cognate Retriever- or Retromer- coats.

Since the FSSS motif of WIPI2 is necessary for its transport functions, we tested whether this motif affects the interactions of WIPI2 with Retriever. To this end, wildtype and the two FSSS variants of $WIPI2^{HA}$ were stably expressed and cell extracts were probed by the same co-adsorption approach as above (Fig. 8). Whereas the Retriever subunit VPS26C was efficiently co-adsorbed with $WIPI2^{WT\text{-}HA}$, $WIPI2^{S67A\text{-}HA}$ and $WIPI2^{S67E\text{-}HA}$ labilized this interaction. Cells carrying only the non-tagged, endogenous WIPI2 yielded no VPS26C signal. Thus, the FSSS motif is necessary for the interaction between WIPI2 and Retriever.

The interaction of WIPI2 and Retriever is necessary for their localization on Rab11-positive recycling endosomes

To test for the in vivo consequences of the WIPI2-Retriever interaction, we analysed the effect of the FSSS motif on the colocalization of the respective $WIPI2^{EGFP}$ variants with different endosomal RAB-GTPases, which were N-terminally tagged with mCherry. In line with previous studies (Polson et al., 2010; Puri et al., 2018) $^{EGFP}WIPI2^{WT}$ mostly co-localized with $^{mCherry}RAB11$ (58%) (Figures 9 and S5-S7) and much less with $^{mCherry}RAB5$ (23%) and $^{mCherry}RAB7$ (6%). By contrast, $^{EGFP}WIPI2^{S67A}$ (63%) and $^{EGFP}WIPI2^{S67E}$ (60%) co-localized mainly with $^{mCherry}RAB5$ (60-63%) and $^{mCherry}RAB7$ (36-38%), and much less with $^{mCherry}RAB11$ (12-14%). The Retriever subunit VPS35L was detected by immuno-staining. It co-localized mainly with $^{mCherry}RAB11$ in cells expressing $WIPI2^{WT}$ (42%) and less with $^{mCherry}RAB5$ (31%) and $^{mCherry}RAB7$ (23%) (Figures. 9, S5-S7). In

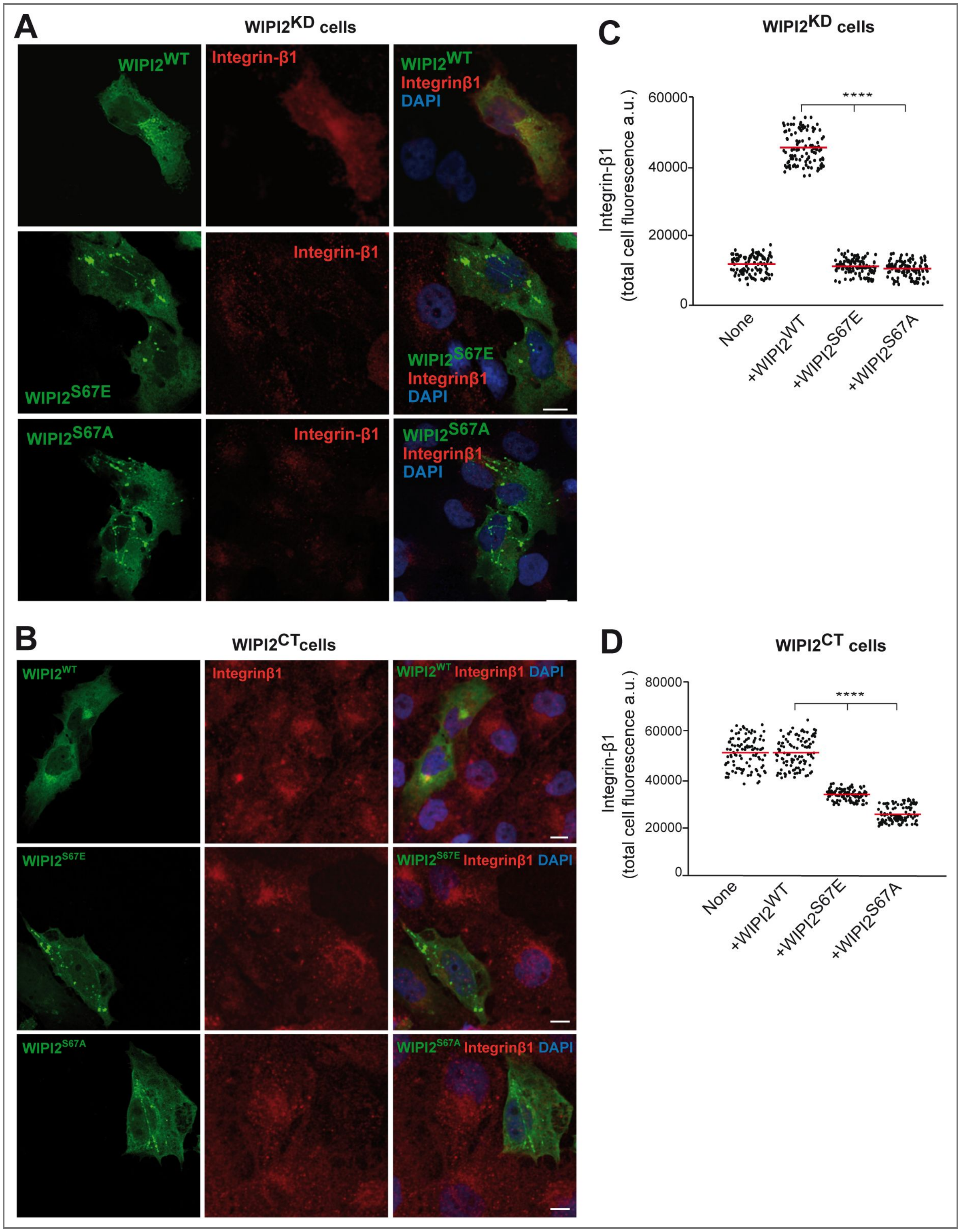

Figure 5. Effects of the WIPI2 FSSS motif on β1-Integrin recycling.

A,B. Influence of WIPI2 variants on β1-Integrin. WIPI2-knockdown (WIPI2^{KD}, A) and control (WIPI2^{CT}) HK2 cells were transfected with a plasmid carrying siRNA-resistant wildtype or the indicated FSSS variants of ^{EGFP}WIPI2. After 18 h of viral transfection, cells were fixed and stained (without detergent permeabilisation) with DAPI and antibodies to β1-Integrin (red). Scale bars: 10 mm. **C,D.** Quantification of β1-Integrin immunofluorescence in cells from A and B. Regions of interest (ROIs) corresponding to cells expressing the indicated WIPI2 variants, and some regions outside the cells (background), were manually defined using Image J software. Total cell fluorescence was integrated and corrected for background fluorescence. 105 cells per condition stemming from three independent experiments were analysed. Red bars indicate the means. P-values were calculated by Welch's t-test. The analysis was performed with 99% confidence: ****P < 0.00001.

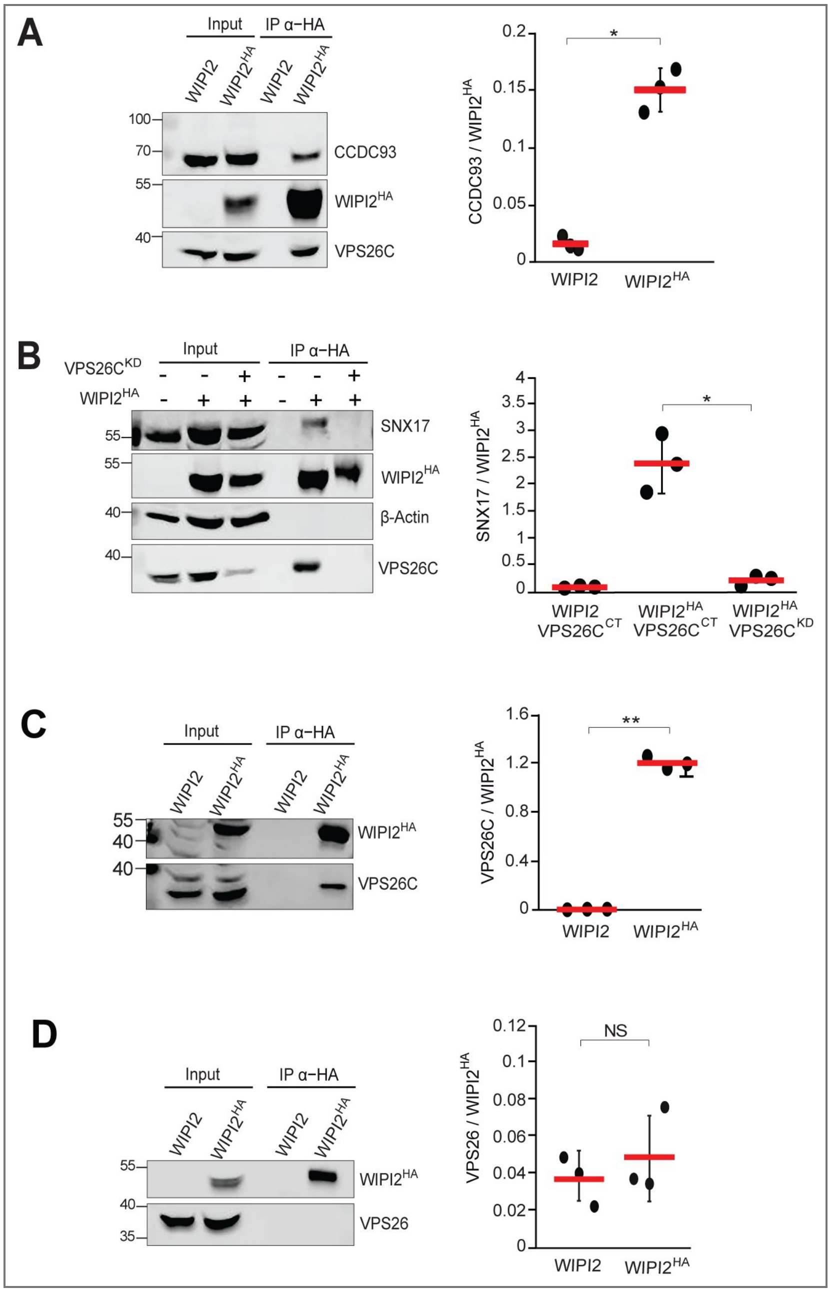

Fig. 6. **Integration of WIPI2 with coat subunits.**

A. Interaction of WIPI2^{HA} with CCDC93. HK2 cells and HK2 cells stably expressing WIPI2^{HA} were detergent solubilised, the total cell extracts were incubated with anti-HA beads and washed. Adsorbed protein was analysed by SDS-PAGE and Western blotting using the indicated antibodies. The intensity of the interacting CCDC93 was quantified with a LICOR Odyssey fluorescence imager. The background from the corresponding position in the sample from cells without HA-tag was subtracted. The resulting intensity is shown relative to the intensity of WIPI2^{HA} signal on the beads. N = 3 biological replicates. Red bars show the means and error bars represent the SEM. P values were calculated with an unpaired Student's t-test. *P < 0.01. **B.** Interaction of WIPI2^{HA} with SNX17. HK2 cells stably expressing WIPI2^{HA} were treated with siRNA to VPS26C (VPS26^{KD}) or with non-specific siRNA (VPS26^{CT}) and lysed. The total cell extracts were incubated with anti-HA beads and adsorbed proteins were analysed using the indicated antibodies as in A. N = 3. **C,D.** Selectivity for Retriever versus Retromer. The co-immunoadsorption experiments were performed as in A and analysed for co-adsorbed (C) VPS26C or (D) VPS26. N=3. *P < 0.01.**P < 0.001. NS=not significant.

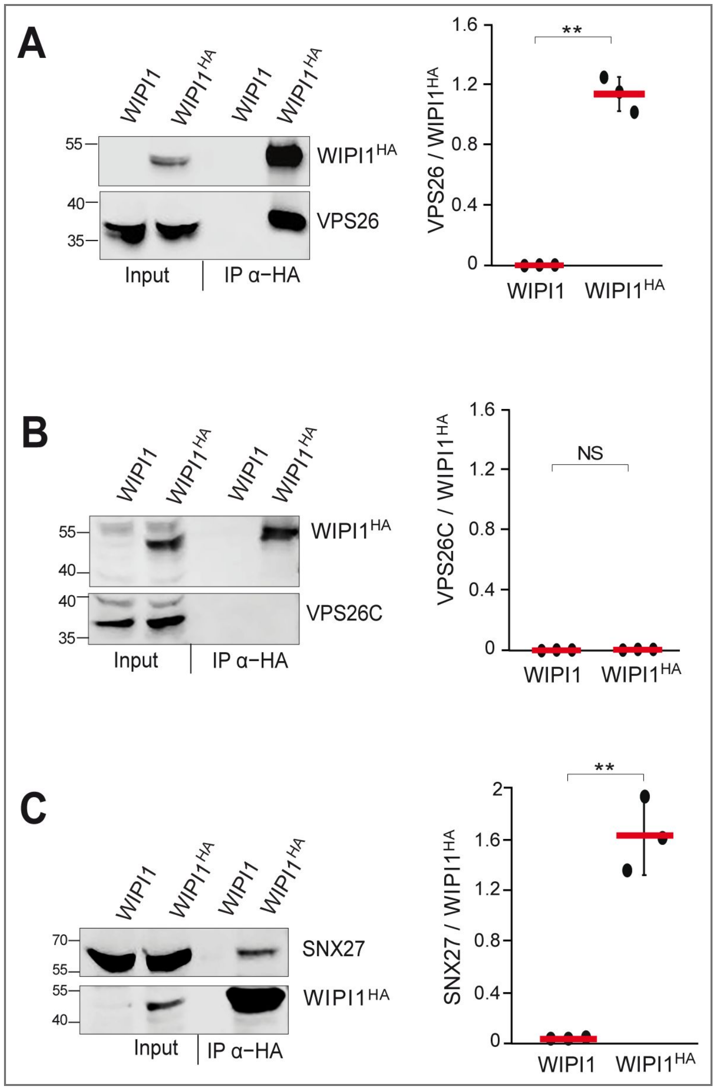

Fig. 7. **Interaction of WIPI1 with VPS26 and SNX27, but not VPS26C.**

A. Interaction of $WIPI1^{HA}$ with VPS26. HK2 cells and HK2 cells stably expressing $WIPI1^{HA}$ were detergent solubilised, the total cell extracts were incubated with anti-HA beads and washed. Adsorbed protein was analysed by SDS-PAGE and Western blotting using the indicated antibodies. The intensity of the interacting VPS26 was quantified with a LICOR Odyssey fluorescence imager. The background from the corresponding position in the sample from cells without HA-tag was subtracted. The resulting intensity is shown relative to the intensity of $WIPI1^{HA}$ signal on the beads. N = 3 biological replicates. Red bars show the means and error bars represent the SEM. P values were calculated with an unpaired Student's t-test. **B.** Lack of interaction of $WIPI1^{HA}$ with VPS26C. HK2 cells stably expressing $WIPI1^{HA}$ were used for co-immunoadsorption experiments as in A and decorated with the indicated antibodies. N = 3. **C**. Interaction of $WIPI1^{HA}$ with SNX27. HK2 cells stably expressing $WIPI1^{HA}$ were used for co-immunoadsorption experiments as in A and decorated with the indicated antibodies. N = 3. **P < 0.001. NS: not significant.

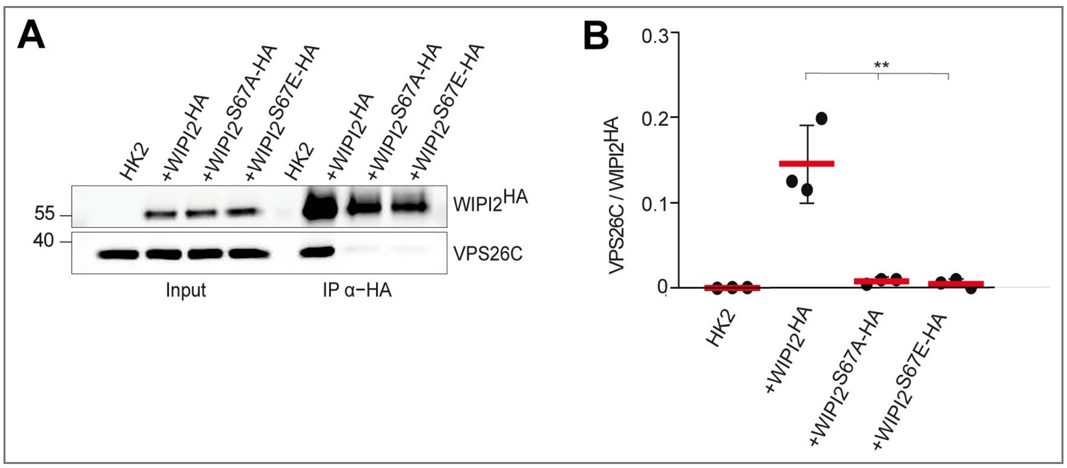

Fig. 8. Impact of the FSSS motif on the WIPI2-Retriever interaction.

A. HK2 cells and HK2 cells stably expressing the indicated $WIPI2^{HA}$ variants were detergent solubilised. Anti-HA beads were incubated with the total cell extracts, washed, and adsorbed proteins were analysed by SDS-PAGE and Western blotting using the indicated antibodies. **B.** Band intensities from the blots in A were quantified with a LICOR Odyssey infrared fluorescence imager and plotted as the ratio of VPS26C over $WIPI2^{HA}$. N=3 independent biological replicates. Red bars show the means and error bars represent the SEM. P values were calculated with an unpaired Student's t-test. **P < 0.001. HC: Heavy chain.

^{EGFP}WIPI2^{S67E} and ^{EGFP}WIPI2^{S67A} expressing cells, VPS35L distribution shifted, similarly as ^{EGFP}WIPI2, from ^{mCherry}RAB11 compartments (12-13%) towards ^{mCherry}RAB5 (42-44%) and ^{mCherry}RAB7 compartments (26-27%). Collectively, these results suggest that the FSSS motif in blade 2 of WIPI2 allows this PROPPIN to associate with Retriever, and that this association promotes re-distribution of WIPI2 and Retriever from Rab5 and Rab7 endosomes to Rab11 compartments.

Discussion

Our experiments reveal a pathway specificity for WIPI1 and WIPI2. The two proteins support distinct cargoes and transport routes and interact with different partners.

This isoform selectivity contrasts with the roles of these proteins in their second functional domain, autophagy, which strongly depends on WIPI2 but is also significantly reduced by ablation of WIPI1 (DeLeo et al., 2021; Dooley et al., 2014; Guan et al., 2001; Obara et al., 2008; Polson et al., 2010; Proikas-Cezanne et al., 2015, 2004). Furthermore, the autophagic function of the PROPPINs also exploits other molecular features than endosomal protein exit. For autophagy, WIPI1 and Atg18 do not require the amphipathic helix in CD loop 6, which is essential for endosomal transport. It was hence proposed that these PROPPINs perform distinct functions in autophagy and in endosomal protein exit, and that their membrane fission activity may only be essential for endosomal transport (Courtellemont et al., 2022; DeLeo et al., 2021; Gopaldass et al., 2017).

In endosomal transport, WIPI1 and WIPI2 display selectivity in their partners. WIPI1 interacts with Retromer and WIPI2 with Retriever, mirroring the impact of the two PROPPINs on the respective transport routes and cargos that they support. WIPI1 and WIPI2 utilise the same molecular features to support these transport routes.

Their FSSS motif is necessary for both isoforms to integrate with Retromer and Retriever, respectively. This, together with the similarity in the overall organisation of Retromer and Retriever, suggest that they might form complexes in a similar manner. In analogy to the WIPI1-Retromer complex, which was termed CROP (Courtellemont et al., 2022), we hence refer to the WIPI2-Retriever complex as CROP2.

WIPI2-dependent transport of β1-Integrin depends also on the amphipathic character of the α-helix in CD loop 6. In Atg18 and WIPI1, this helix is necessary to convey membrane fission activity, which is used both for division of the lysosome-like vacuoles of yeast and for detachment of endosomal carriers (DeLeo et al., 2021; Gopaldass et al., 2017; Zieger and Mayer, 2012). This fission activity is potentiated by integrating Atg18 and WIPI1 with Retromer into the CROP complex (Courtellemont et al., 2022). Our observations suggest that WIPI2 contributes pathway-specific membrane fission activity by integrating with Retriever. In line with this, ablating the amphipathic character of its helix on CD loop 6 disrupts β1-Integrin recycling to the cell surface and leads to its accumulation in endosomal compartments. Disrupting the WIPI2-Retriver interaction by substitution of the FSSS motif has the same effect. It also leads to the accumulation of long tubular structures, which we presume to be exaggerated endosomal carriers that continue to grow but fail to detach. This phenotype is consistent with a block in the membrane fission step that is necessary to allow the endosomal carrier to depart.

Retriever couples to other proteins that are necessary for endosomal protein transport. These include the CCC complex, which associates with Retriever to yield the very large Commander complex (Boesch et al., 2023b; Healy et al., 2023; Laulumaa et al., 2024). How Commander is placed on the membrane is currently unknown, but the interaction between SNX17 and Retriever contributes to its recruitment (Butkovič et al., 2024; Healy et al., 2022; Singla et al., 2024; Steinberg et al., 2012) and presumably to the formation of a coat. Our pull-downs of WIPI2 detected not only the Retriever subunit VPS26C as an interactor, but also SNX17 and the CCC subunit CCDC93. This suggests that WIPI2 integrates into the coat and might thus bring membrane fission activity to the place where it is required for carrier formation. Here, it could synergize with

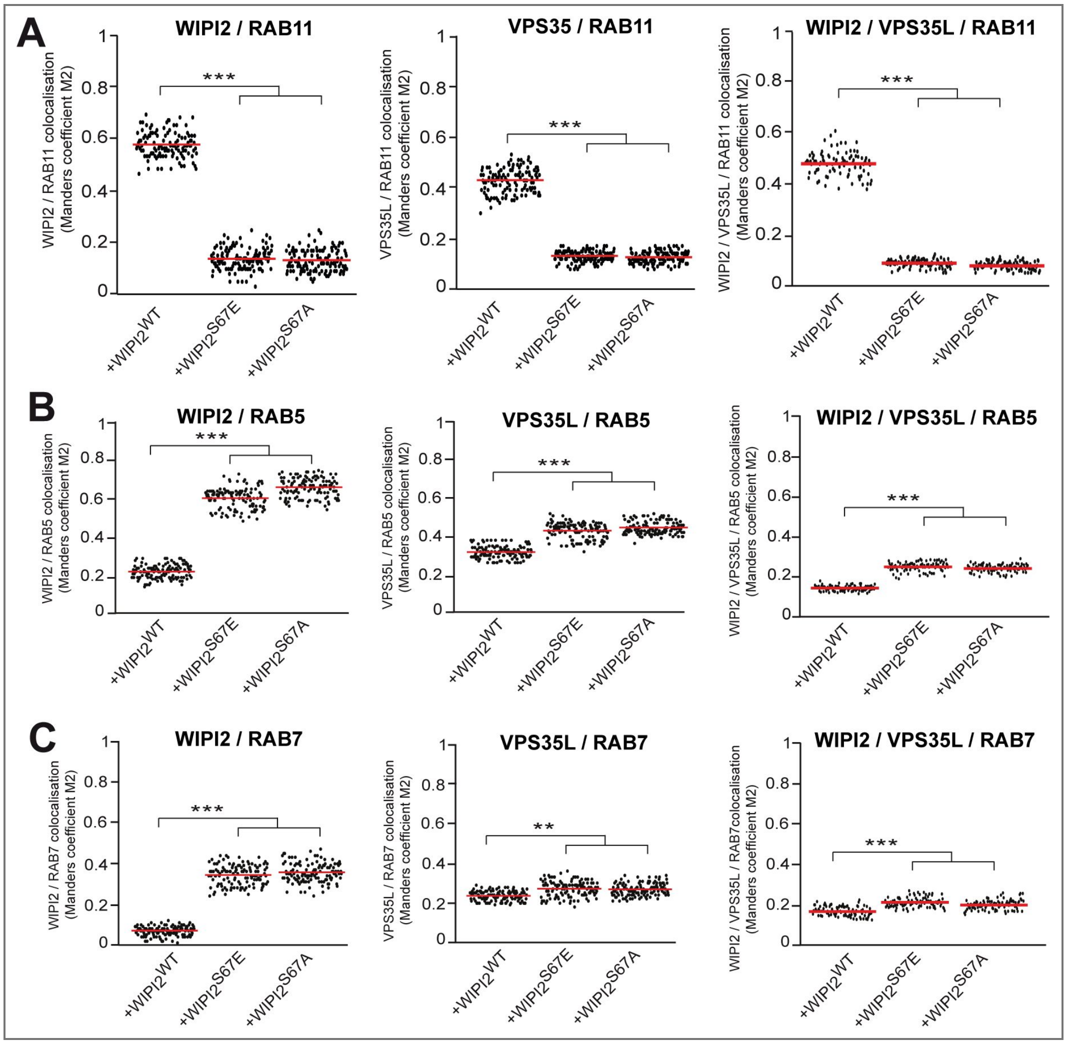

Figure 9. Role of the WIPI2 FSSS motif for recruiting WIPI2 and Retriever to Rab11 endosomes.

WIPI2-knockdown HK2 cells were transfected with plasmids expressing the indicated siRNA- resistant ^{EGFP}WIPI2 variants and mCherry-RAB5, mCherry-RAB7, or DsRed-Rab11 were fixed, permeabilized, immuno-stained for VPS35L and analyzed by confocal microscopy. Examples of the quantified images are presented in Supplementary Figures S5 to S7. Colocalization with ^{EGFP}WIPI2 and VPS35L was assessed for: **A.** RAB11 **B.** RAB5 **C.** RAB7. Colocalization was quantified using Manders' colocalization coefficient M2, calculated in ImageJ. M2 refers to the fraction of VPS35L colocalising with the different Rab-proteins. For the triple colocalizations WIPI2/VPS35L/RAB M2 indicates the fraction of green pixels (WIPI2) overlapping with the pixels positive for the VPS35L/RAB colocalization. Colocalization was quantified from three independent experiments in a total of 120 cells. P values were calculated applying Welch's t-test with unequal variances. The analysis was performed with 99% confidence. ***P < 0.0001.

other factors that support membrane fission in endosomal transport, such as EHD1, which interacts with SNX17, and the actin- regulating WASH complex, which binds to Commander (Bartuzi et al., 2016; Dhawan et al., 2022, 2020; Phillips-Krawczak et al., 2015).

The requirement of two distinct WIPI proteins, using the same molecular features, in Retromer- and Retriever-dependent transport uncovers a novel conserved element between these two pathways. In combination with the similar overall organisation of Retromer and Retriever and their common interactions with EHD proteins (Arlt et al., 2015; Chi et al., 2014; Daumke et al., 2007; Deo et al., 2018; Gokool et al., 2007; Grant et al., 2001; Solinger et al., 2020) and WASH (Derivery et al., 2009; Gomez and Billadeau, 2009; Guo et al., 2024; Harbour et al., 2012; Jia et al., 2012; Phillips-Krawczak et al., 2015; Seaman et al., 2009), this supports the notion that these two membrane coats share similar mechanistic principles for their formation and/or detachment.

Materials and Methods

Antibodies and reagents

All chemical reagents were from Sigma-Aldrich unless specified otherwise

We used anti-LAMP1 (H4A3, US Biological, Life Sciences); anti- EEA1 (#2411, Cell Signaling); anti-EGFR (sc-373746, clone A-10 Santa Cruz Biotechnology; PA5- 85089 ThermoFisher Scientific); anti-Integrin β1 (ab150361, Abcam); anti-GLUT1 (ab15309, Abcam); anti-c16orf62/VPS35L (PA5-28553, ThermoFisher Scientific); anti-SNX17(HPA043867, Atlas Antibodies); anti-SNX27 (ab77799, Abcam); anti CCDC93 (20861-1AP, Proteintech); anti-α-Tubulin (T9026, Sigma); anti-Vps26 (ab181352, Abcam); anti-VPS26C/DSCR3 (ABN87, Merck Millipore), WIPI1 (W2394, Sigma); WIPI2 (HPA019852, Sigma).

As secondary antibody we used: Cy3-conjugated Affinity Pure Donkey anti-Mouse IgG H + L (Jackson Immuno Research); Cy3-conjugated AffiniPure Donkey anti- Rabbit IgG H + L (Jackson Immuno Research); Alexa Fluor®488-conjugated AffiniPure Donkey anti-Rabbit IgG H + L (Jackson Immuno Research); Alexa Fluor®488-conjugated AffiniPure Donkey anti-Mouse IgG H + L (Jackson Immuno Research); Alexa Fluor® 647-conjugated AffiniPure donkey anti-rabbit IgG H + L (Jackson Immuno Research); Alexa Fluor® 647-conjugated AffiniPure donkey anti- Mouse IgG H + L (Jackson Immuno Research). Antibodies were used at dilutions summarized in Supplementary Table1.

Other reagents: Opti-MEM (Thermo Fischer, 11058021) and Trypsin (Thermo Fischer, 27250018).

Protease inhibitor (PI) cocktail final concentrations in samples: 100 µM pefablock SC (Merck, 11429876001), 2 µM leupeptin (Merck, 11529048001), 50 µM o-phenanthroline (Merck ,131377), 0.7 µM pepstatin A (Merck, 11524488001) dissolved in methanol.

Phosphatase inhibitor cocktail final concentrations: 10 mM NaF (S-7920, Sigma); 1mM Na Orthovanadate (S6508, Sigma); 2 mM β-glycerophosphate (A2253, AppliChem) all dissolved in water.

Complementary DNA (cDNA) constructs

Vectors expressing tagged RAB proteins were purchased from Addgene: DsRed- RAB11 (12679; deposited by R. Pagano); mCherry-RAB7 (61804; deposited by G. Voeltz); mCherry-RAB5 (49201; deposited by G. Voeltz). The following vectors were kindly provided by colleagues: EGFP-WIPI2 (pAR31CD vector) (Tassula Proikas- Cezanne, Tübingen, Germany). The sequences of WIPI1 and WIPI2 that have been used are given in Supplementary File 1.

Site-directed mutagenesis

All constructs were verified by DNA sequencing. ^{EGFP}WIPI2 or WIPI2^{HA} was used as DNA template for site-directed mutagenesis (QuikChange mutagenesis system, Agilent Technologies, 200524) to generate point mutations in the FSSS motif (S67A and S67E) following the manufacturer's protocol.

To get WIPI2^{S67E} we used the following primers: Fw: AGATTGTTCTCCGAGAGCCTAGTGGCC and Rv GGC CAC TAG GCT CTC GGA

GAA CAA TCT; while for WIPI2^{S67A} we used Fw: AGATTGTTCTCCGCTAGCCTAGTGGCC

Rv: GGCCACTAGGCTAGCGGAGAACAATCT (Microsynth).

WIPI2^{SLoop} was generated by four consecutive amplifications using the following primers:

1. Fw: TGGACC GGGTACAAAGGGAAAGTG CTC Rv: GAGCACTTTCCCTTTGTACCCGGTCCA;
2. Fw: GGGTACAAAGGGTTCGTGCTCATGGCC Rv: GGCCATGAGCACGAACCCTTTGTACCC;
3. Fw: CTC ATG GCC TCC TAC AGC TAC CTG CCT Rv: AGGCAGGTAGCTGTAGGAGGCCATGAG;
4. Fw: GCC TCC TAC AGC ACC CTG CCT TCC CAA Rv: TTGGGAAGGCAGGGTGCTGTAGGAGGC

Non-mutated template vector was removed from the PCR mixture through digestion by the enzyme Dpn1 for 1 h at 37°C. The product was purified using NucleoSpin PCR Clean-up (Macherey-Nagel, 740609.50S) and transformed into *Escherichia coli*. Plasmid DNA was purified and sequenced. The site-directed mutagenesis system was also used to generate the siRNA-resistant cDNA for WIPI2 with primers Fw: CGATAGTCCTTTAGCCGCA and Rv: TGCGGCTAAAGGACTATCG. All oligonucleotides were synthesized by Microsynth.

Cell culture, transfection and treatments

HK2 cells were grown in DMEM-HAM's F12 (Thermo Fischer, 11765054) supplemented with 5% fetal calf serum (Gibco, 10270106), 50 U/mL penicillin/50 mg/mL streptomycin (ThermoFischer, 15140148), ITS (5 µg/mL insulin, 5 µg/mL TF, 5 ng/mL selenium; LuBio Science, 00-101-100ML). Cells were grown at 37°C in 5% CO2 and at 98% humidity.

HK2 cells were transfected with plasmids using the X-tremeGENE^{TM} HP DNA transfection reagent (Sigma-Aldrich, 6366546001), unless otherwise specified, according to the manufacturer's instructions, and incubated for 18-24 h before fixation or live cell imaging. The HK2 cell line was checked for mycoplasma contamination by a PCR-based method. All cell-based experiments were independently repeated at least three times.

Generation of HK2 cells expressing WIPI2^{HA}

HK2 cell lines stably expressing the three forms (WT, S67A, S67E) of C-terminal HA- tagged WIPI2 were generated by lentiviral transduction. WIPI2 was cloned into modified pLKO.1 lentiviral vector (available from Sigma) between Xba1 and EcoR1 restriction sites. To obtain WIPI2^{HA} a sequence of 3Gly - 1Ser - 3Gly - 1Ser - HA - HA (designed and purchased from GeneScript) was added using Gibson assembly according to the manufacturer's protocol. Finally, the lentiviral plasmid DNA was purified and sequenced. Lentiviral transfer vector together with third generation envelope and packaging plasmids were transfected into 293T cells to generate lentivirus. Packaged lentivirus containing supernatant was added to recipient HK2 cells in the presence of 10 µg/ml Polybrene (AL-118, Sigma-Aldrich). HK2 cells were stably infected with a lentiviral vector for constitutive WIPI2^{HA} expression for 6 h. 2 days after infection, cells were selected with 1 µg/ml of puromycin for 3 days. Protein samples were collected and used for IP experiments 7 days after infection.

RNA interference

HK2 cells were transfected with siRNA for 72 h using Lipofectamine® RNAiMax (Thermo Fisher Scientific, 13778150) according to the manufacturer's instructions. Control cells were treated with identical concentrations of siGENOME Control Pool Non-Targeting from Dharmacon (D-001206-13-05). siRNAs targeting WIPI2 were from Dharmacon (ON-TARGETplus Human WIPI2 J-020521). siRNAs were used at a final concentration of 20 nM.

Endocytosis assays

EGFR degradation

HK2 cells were seeded in DMEM-HAM's F12 supplemented with 5% foetal calf serum, 50 U/mL penicillin/50 mg/mL streptomycin and ITS (5 µg/mL insulin, 5 µg/mL TF, 5 ng/mL selenium. The day after the growth medium was replaced with the same but without serum for 24 h and HK2 cells were stimulated with 100 ng/ml EGF for the indicated periods of time. The cells were fixed for 10 min in 4% paraformaldehyde in PBS at different time points after stimulation.

Immunofluorescence

Cells were fixed for 10 min in 4% paraformaldehyde in PBS. After fixation, cells were permeabilized in 0.05% (w:v) saponin (Sigma-Aldrich, 558255), 0.5% (w:v) BSA and 50 mM NH4Cl in PBS (blocking buffer) for 30 min at room temperature. The cells were incubated for 1 h with primary antibodies in blocking buffer, washed three times in PBS, incubated for 1 h with the secondary antibodies (Alexa Fluor-conjugated), washed three times in PBS, mounted with fluorescence mounting medium (Dako, S3023) on slides and analysed by confocal microscopy.

Confocal fluorescence microscopy, image processing, and colocalization analysis

HK2 cells were grown to 70% confluence on glass coverslips and immunofluorescence microscopy was performed as described above. The experiments were repeated at least three times and representative images are shown. Colocalization was analysed by acquiring serial sections from about 100 cells per sample. Images were exported in TIFF format and processed as previously described (Vicinanza et al., 2011). Images of samples to be compared were acquired using the same settings (i.e. laser power, photomultiplier gain and pinhole size), avoiding pixel saturation. The images were processed in the same way using ImageJ software. Channels from each image were converted into 8-bit format and the 'Auto Local Threshold' Plug-in with the 'Default' method was used to segment grayscale images, identify the structures of interest and subtract background.

GLUT1 and β1-Integrin fluorescence levels were quantified in CTR and *WIPI2 KD* cells using ImageJ. *z*-stack images were acquired and compressed into a single plane using the 'maximum intensity Z-projection' function in Image J. Individual cells were selected using the freeform drawing tool to create a ROI (ROI). The 'Measure' function provided the area, the mean grey value and integrated intensity of the ROI. The mean background level was obtained by measuring the intensity in three different regions outside the cells, dividing them by the area of the regions measured, and averaging the values obtained. The background was subtracted from each cell to obtain the CTCF (corrected total cell fluorescence). It was calculated using the formula: CTCF=integrated intensity of cell ROI − (area of ROI × mean fluorescence of background).

To quantify the degree of colocalization, confocal z-stacks were acquired, single channels from each image in 8-bit format were thresholded to subtract background and then the "Just Another Colocalisation Plug-in" (JACOP) was used to measure the overlap coefficient according to Manders. Manders' coefficient indicates the overlap of the signals and represents the degree of colocalization:

$$R = \frac{\sum_i S1_i \cdot S2_1}{\sqrt{\sum_i (S1_i)^2 \cdot \sum_i (S2_1)^2}}$$

where $S1$ represents signal intensity of pixels in the channel 1 and $S2$ represents signal intensity of pixels in the channel 2. The colocalization coefficients M1 and M2 indicate the contribution of each channel to the pixels. They are independent of the intensity of the overlapping pixels, but they are sensitive to background, so a threshold must be set (Manders et al., 2011). In our colocalization analysis, M1 indicates the fraction of red pixels overlapping with green one, while

M2 indicates the fraction of green pixels overlapping with the red ones. Confocal microscopy was performed on an inverted confocal laser microscope (Zeiss LSM 880 with airyscan) with a 63x 1.4 NA oil immersion lens unless stated otherwise.

Gel electrophoresis and Western blot

Cells were plated into 12-well tissue culture test plates (TPP) until they reached 90% confluency, except for knockdown-cells, which were cultured until 72h after transfection with the siRNAs. Cells were then washed three times with ice-cold PBS (phosphate-buffered saline), scraped, and proteins were extracted in ice-cold lysis buffer (150 mM NaCl, 2 mM EDTA, 40 mM HEPES-NaOH pH 7.4, and 1% Triton X- 100 supplemented with phosphatases and protease inhibitor cocktail (see details in Reagent section). Protein extracts were supplemented with 1/4 volume of 5x reducing sample buffer (250 mM Tris-Cl, pH 6.8, 5% β-mercaptoethanol, 10% SDS, 30% glycerol, 0.02% bromophenol blue) and heated to 95 °C for 5 min. The samples were run on either 8%, 10% or 12.5% SDS-polyacrylamide gels (W x L x H: 8.6 x 6.8 x 0.15 cm). The stacking gels were prepared as follows: 6% acrylamide, 0.16% bis- acrylamide, 0.1 M Tris, pH 6.8, 0.1% SDS, 0.1% TEMED, 0.05% ammonium persulfate. Running gels were: 10% or 12.5% acrylamide, 0.27% or 0.34% bis-acrylamide, 0.38 M Tris, pH 8.8, 0.1% SDS (Applichem, 475904-M), 0.06% TEMED (Applichem, A1148), 0.06% APS (Applichem, A2941). The gels were run at constant current (20–30 mA). Proteins were blotted onto nitrocellulose membrane by the semi-dry method for 80 min at 400 mA (Trans-Blot® SD Semi-Dry Electrophoretic Transfer Cell, Bio-Rad). After incubation with the primary antibody, signals were detected by secondary antibodies coupled to infrared dyes (LI-COR) and detected on a LI-COR Odyssey infrared fluorescence imager. Images were exported as TIFF files and processed in Adobe Photoshop. Band intensity was quantified using ImageJ band analysis.

Immunoprecipitation of Retriever

HK2 cells expressing the three variants (WT, S67E, S67A) of WIPI2 tagged with HA were grown to 80% confluence in 15 cm dishes. After rinsing the cells three times with phosphate buffered saline (PBS 1X), they were detached and lysed by incubating 15 min at 4°C with lysis buffer: 150 mM NaCl, 2 mM EDTA, 40 mM HEPES-NaOH pH 7.4, 1% Triton X-100, supplemented with phosphatases and proteases inhibitor cocktails (details in Reagent section) and 1 mM PMSF out of a 200 mM solution in ethanol, which was freshly prepared immediately before use. The lysate was spun for 10 min at 10,000 x g on a cold bench-top centrifuge). 2.5% of the clarified extracts was reserved as “input”. Each clarified extract was incubated for 2 h at 4°C with 20 µl of Anti-HA High Affinity antibody (clone 3F10) (Roche #118150116001). Then, samples were washed three times with lysis buffer and, after discarding the last wash, the beads were resuspended in 50 µl of pre-warmed elution buffer (2% SDS, 1mM EDTA, 10mM DTT, 20mM Tris HCl pH 7.4, 100 µM PMSF) and shaken at 55°C for 10 min. Samples were centrifuged, the supernatants were transferred to new Eppendorf tubes, and 5x sample buffer containing 100 mM of fresh DTT was added. Samples were kept at 95°C for 5 min and centrifuged in a table to centrifuge. All supernatants were transferred to a new Eppendorf tube yielding the “IP” samples. Inputs and IP were loaded on 8% or 10% SDS–polyacrylamide gels.

Statistical analysis

Differences between the means have been evaluated by two-sample Student’s t-test assuming unequal variances unless otherwise specified. They are mentioned as significant differences when $p < 0.01$. Immunofluorescence experiments were independently repeated three times and at least 35 cells were analysed from a single experiment. Representative images are shown. Western blotting experiments were independently repeated at least three times and representative blots are shown. Most data are presented as the means ± standard deviation (s.d.) unless otherwise specified.

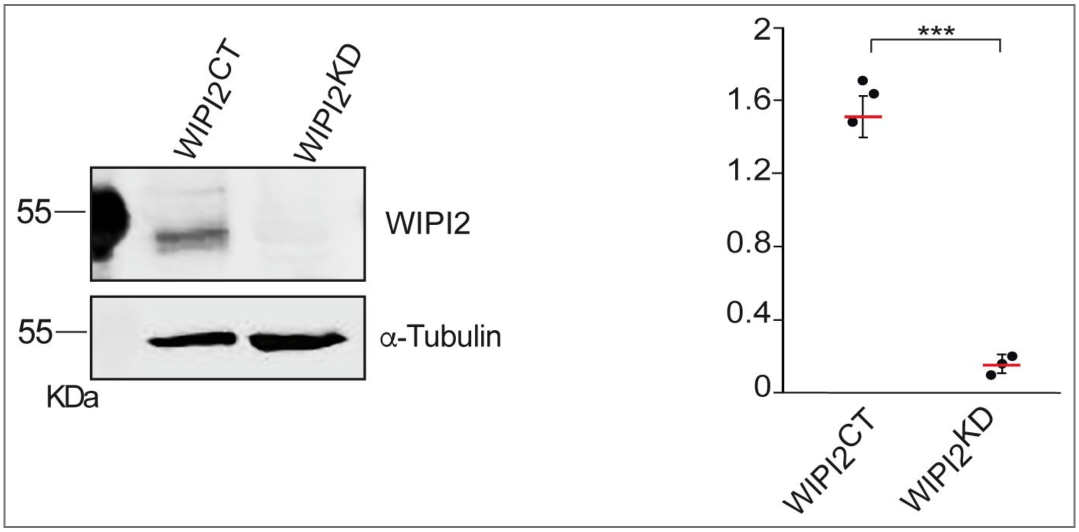

Fig. S1. Efficiency of WIPI2 knockdown. Lysates of HK2 cells (50 µg per sample) were treated with non-target siRNA (WIPI2^{CT}) or siRNA against WIPI2 (WIPI2^{KD}). HK2 cells were analyzed by SDS-PAGE and Western blot against the indicated proteins. a-tubulin was used as a loading control. A representative blot is shown. The signals were quantified on a LICOR Odyssey infrared fluorescence image WIPI2/a-Tubulin ratios were calculated. Red bars indicate the mean and error bars the SEM; n=3 independent experiments. P-values were calculated by Welch's t-test. ***P < 0.0001.

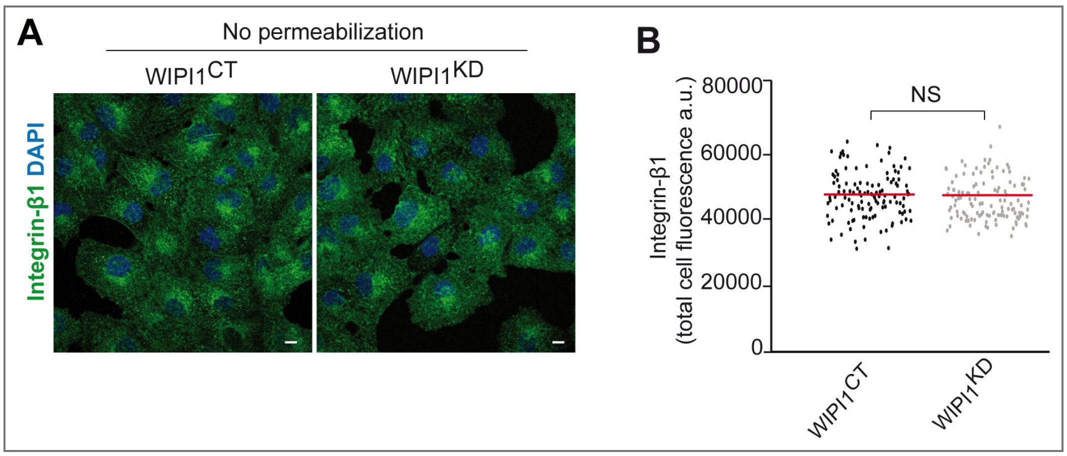

Figure S2. **Surface expression of β1-Integrin in cells depleted of WIPI1.**

A WIPI1^{CT} and WIPI1^{KD} cells were fixed and stained with DAPI and with antibody to β1-Integrin. Scale bars: 10 µm. **B** Quantification of β1-Integrin-immunofluorescence in cells from A. Regions of interest (ROIs) corresponding to each cell and in some regions outside the cells (background) were manually defined using ImageJ software. Total cell fluorescence was integrated and corrected for background fluorescence. 150 cells per condition, stemming from three independent experiments, were analyzed. Red bars show the means. P values were calculated by unpaired Student's t-test. The analysis was performed with 99% confidence. NS = not significant (P>0.05).

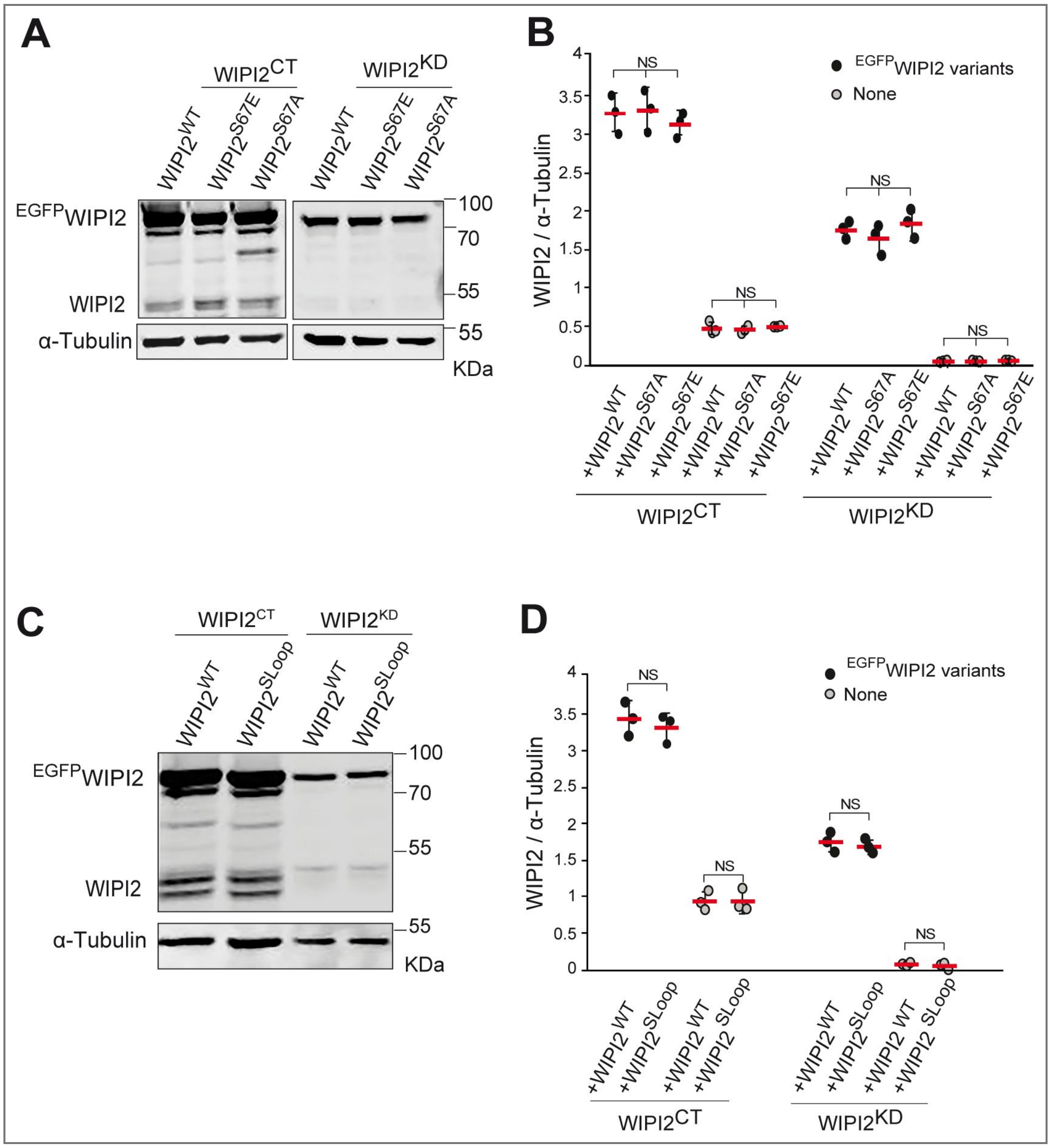

Figure S3. Expression levels of WIPI2 variants.

A. FSSS variants. Cell lysates from wild-type and WIPI2^{KD} cells were transfected for 18 h with siRNA resistant lentiviral expression constructs for ^{EGFP}WIPI2^{WT}, ^{EGFP}WIPI2^{S67A} and ^{EGFP}WIPI2^{S67E}. The cells were analyzed by SDS–PAGE and Western blotting with antibodies to WIPI2 and α-Tubulin. A representative blot is shown. **B** Quantification of the signals for virally expressed ^{EGFP}WIPI2 variants and for endogenous WIPI2. Blots from three independent experiments were quantified on a Licor Odyssey infrared scanner. Graphs on the right side show the mean and SEM. P-values were calculated by Welch's t-test. NS: not significant. **C.** SLoop variant. ^{EGFP}WIPI2 and ^{EGFP}WIPI2^{SLoop} were expressed and analyzed by Western blotting as in described in B. **D** Blots from C were quantified as in B.

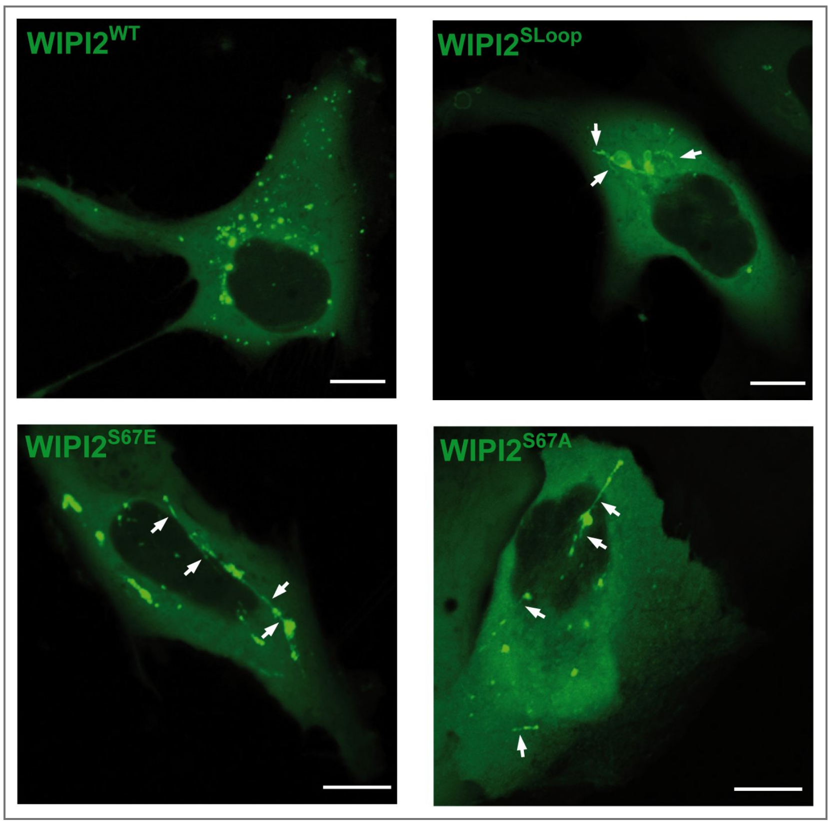

Figure S4. **Live cell imaging of cells expressing ^{EGFP}WIPI2 variants**

WIPI2^{KD} cells were transfected with plasmids carrying the indicated siRNA-resistant ^{EGFP}WIPI2 variants. After 18h of viral transfection, cells were analysed by live cell confocal microscopy. Arrows point to tubular structures that are readily detectable in cells expressing the SLoop and S67 variants of WIPI2. Scale bars: 10 mm.

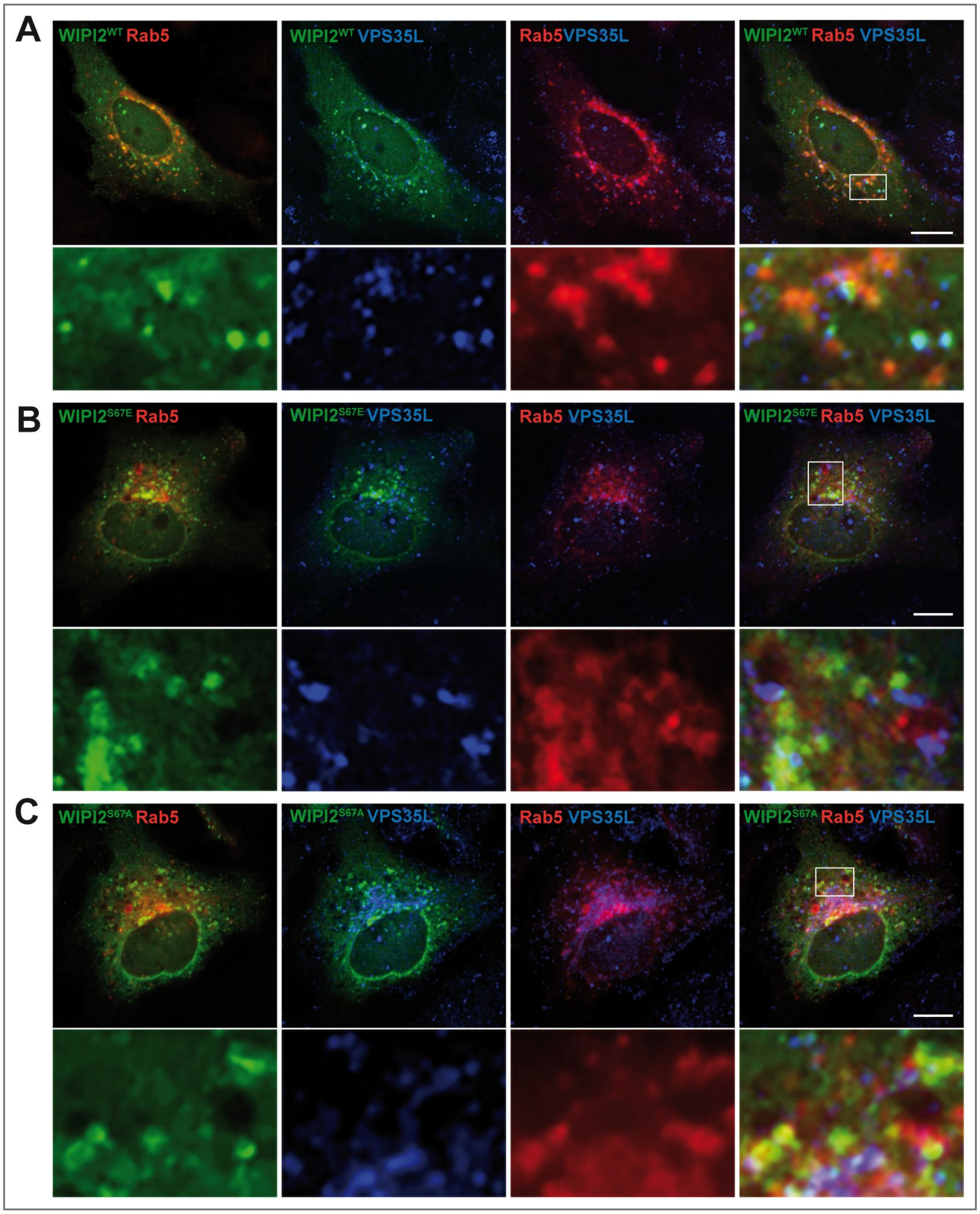

Figure S5. WIPI2 colocalization with RAB5 and Retriever.

HK2 cells expressing mCherry-RAB5 and ^{EGFP}WIPI2 wild-type (**A**) or its indicated variants (**B-C**), were fixed, permeabilized and stained with antibodies to VPS35L (blue). Scale bars: 10 μm. Insets show enlargements of the outlined areas. Quantifications from multiple experiments are shown in Fig. 9

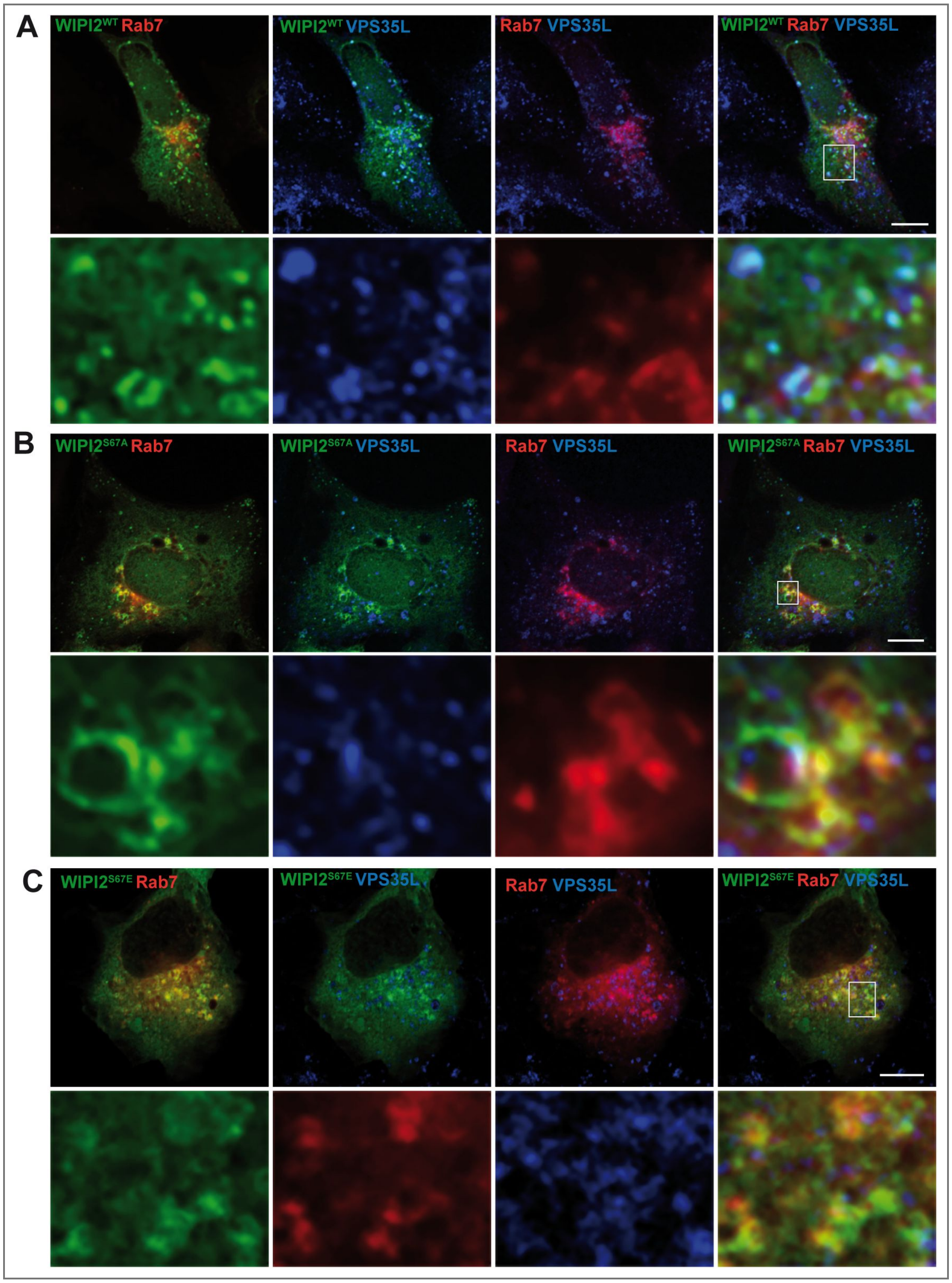

Figure S6. WIPI2 colocalization with RAB7 and Retriever.

HK2 cells expressing mCherry-RAB7 and wild-type ^{EGFP}WIPI2 (**A**) or its indicated variants (**B-C**) were fixed, permeabilized and stained with antibodies to VPS35L (blue). Scale bars: 10 µm. Insets show enlargements of the outlined areas. Quantifications from multiple experiments are shown in Fig. 9

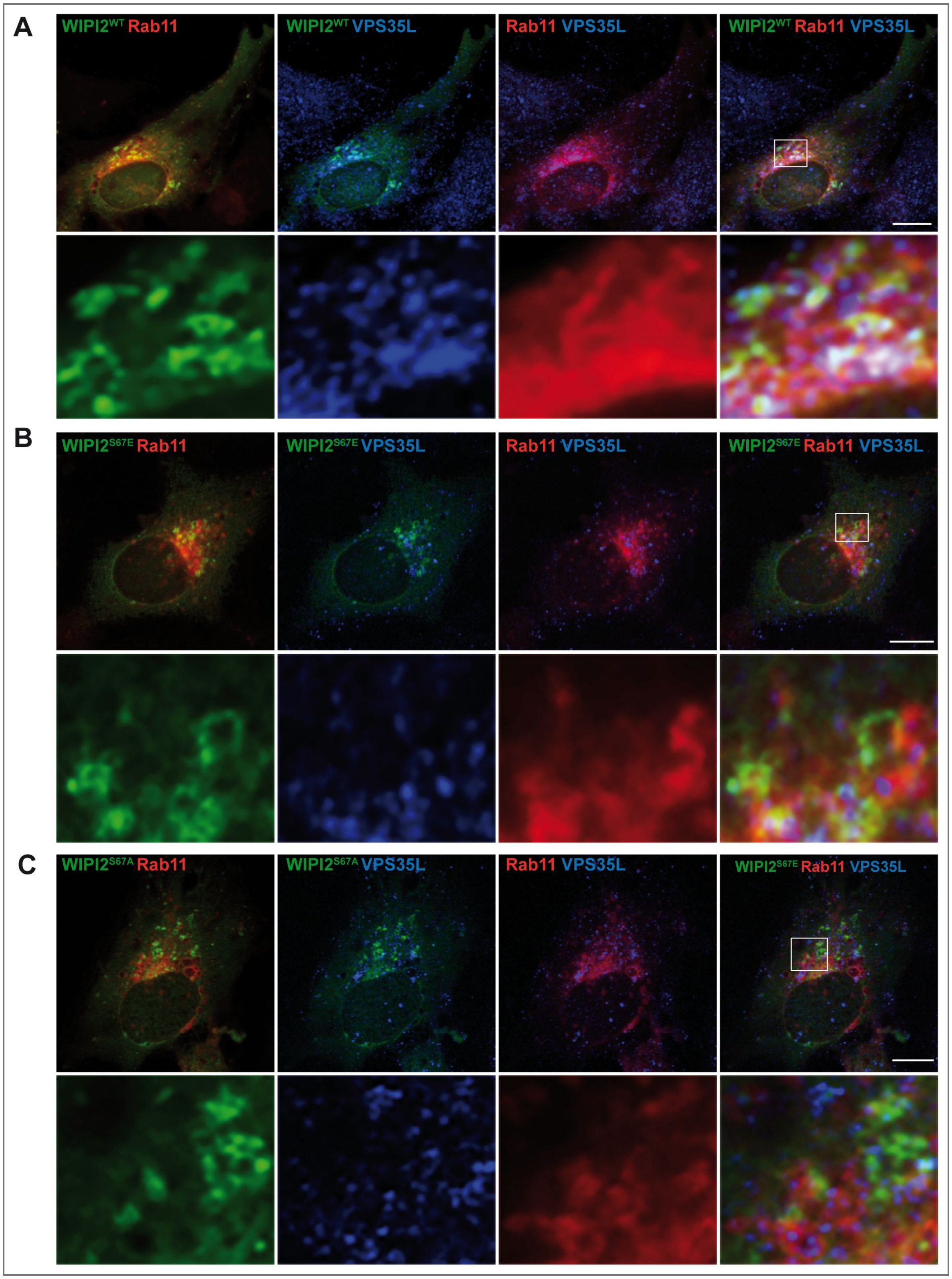

Figure S7. WIPI2 colocalization with RAB11 and Retriever.

HK2 cells expressing mCherry-RAB7 and wild-type ^{EGFP}WIPI2 (**A**) or its indicated variants (**B-C**) were fixed, permeabilized and stained with antibodies to VPS35L (blue). Scale bars: 10 µm. Insets show enlargements of the outlined areas. Quantifications from multiple experiments are shown in Fig. 9

Data availability

Primary data, clonal reagents and cell lines will be supplied by the authors upon request.

Acknowledgements

We thank Navin Gopaldass and Thibault Courtellemont for discussions. This work was supported by grants from the SNSF (31003A_179306) and ERC (788442) to AM.

Additional files

Supplementary File 1

Supplementary File 2

Additional information

Funding

Funder	Grant reference number	Author
Schweizerischer Nationalfonds zur Förderung der Wissenschaftlichen Forschung (SNF)	31003A_179306	Andreas Mayer
EC | European Research Council (ERC)	https://doi.org/10.3030/788442	Andreas Mayer

Author ORCID iDs

Andreas Mayer: http://orcid.org/0000-0001-6131-313X

References

1. **Antonny B,** Burd C, Camilli PD, Chen E, Daumke O, Faelber K, Ford M, Frolov VA, Frost A, Hinshaw JE, *et al.* (2016) Membrane fission by dynamin: what we know and what we need to know. *EMBO J* **35**:2270-2284 https://doi.org/10.15252/embj.201694613
2. **Arlt H,** Reggiori F, Ungermann C (2015) Retromer and the dynamin Vps1 cooperate in the retrieval of transmembrane proteins from vacuoles. *J Cell Sci* **128**:645-655 https://doi.org/10.1242/jcs.132720
3. **Bakula D,** Takacs Z, Proikas-Cezanne T (2013) WIPI beta-propellers in autophagy- related diseases and longevity. *Biochem Soc Trans* **41**:962-7 https://doi.org/10.1042/bst20130039
4. **Bartuzi P,** Billadeau DD, Favier R, Rong S, Dekker D, Fedoseienko A, Fieten H, Wijers M, Levels JH, Huijkman N, *et al.* (2016) CCC- and WASH- mediated endosomal sorting of LDLR is required for normal clearance of circulating LDL. *Nat Commun* **7**:10961 https://doi.org/10.1038/ncomms10961
5. **Baskaran S,** Ragusa MJ, Boura E, Hurley JH (2012) Two-Site Recognition of Phosphatidylinositol 3-Phosphate by PROPPINs in Autophagy. *Mol Cell* **47**:339-348 https://doi.org/10.1016/j.molcel.2012.05.027
6. **Bean BD,** Davey M, Conibear E (2017) Cargo selectivity of yeast sorting nexins. *Traffic* **18**:110-122 https://doi.org/10.1111/tra.12459
7. **Boesch DJ,** Singla A, Han Y, Kramer DA, Liu Q, Suzuki K, Juneja P, Zhao X, Long X, Medlyn MJ, *et al.* (2023a) Structural Organization of the Retriever-CCC Endosomal Recycling Complex. *bioRxiv* https://doi.org/10.1101/2023.06.06.543888
8. **Boesch DJ,** Singla A, Han Y, Kramer DA, Liu Q, Suzuki K, Juneja P, Zhao X, Long X, Medlyn MJ, *et al.* (2023b) Structural organization of the retriever–CCC endosomal recycling complex. *Nat Struct Mol Biol* **31**:910-924 https://doi.org/10.1038/s41594-023-01184-4

9. **Busse RA,** Scacioc A, Krick R, Pérez-Lara Á, Thumm M, Kühnel K (2015) Characterization of PROPPIN-Phosphoinositide Binding and Role of Loop 6CD in PROPPIN-Membrane Binding. *Biophys J* **108**:2223-2234 https://doi.org/10.1016/j.bpj.2015.03.045

10. **Butkovič R,** Walker AP, Healy MD, McNally KE, Liu M, Veenendaal T, Kato K, Liv N, Klumperman J, Collins BM, *et al.* (2024) Mechanism and regulation of cargo entry into the Commander endosomal recycling pathway. *Nat Commun* **15**:7180 https://doi.org/10.1038/s41467-024-50971-0

11. **Carosi JM,** Denton D, Kumar S, Sargeant TJ (2023) Receptor Recycling by Retromer. *Mol Cell Biol* **2023;43**:317-334 https://doi.org/10.1080/10985549.2023.2222053

12. **Chandra M,** Kendall AK, Ford MGJ, Jackson LP. (2024) VARP binds SNX27 to promote endosomal supercomplex formation on membranes. *bioRxiv* https://doi.org/10.1101/2024.07.11.603126

13. **Chi RJ,** Liu J, West M, Wang J, Odorizzi G, Burd CG (2014) Fission of SNX-BAR– coated endosomal retrograde transport carriers is promoted by the dynamin- related protein Vps1. *J Cell Biol* **204**:793-806 https://doi.org/10.1083/jcb.201309084

14. **Clippinger AK,** Naismith TV, Yoo W, Jansen S, Kast DJ, Hanson PI (2024) IST1 regulates select recycling pathways. *Traffic* **25**:e12921 https://doi.org/10.1111/tra.12921

15. **Collins BM,** Norwood SJ, Kerr MC, Mahony D, Seaman MNJ, Teasdale RD, Owen DJ (2008) Structure of Vps26B and mapping of its interaction with the retromer protein complex. *Traffic* **9**:366-379 https://doi.org/10.1111/j.1600-0854.2007.00688.x

16. **Cong Y,** So V, Tijssen MAJ, Verbeek DS, Reggiori F, Mauthe M (2021) WDR45, one gene associated with multiple neurodevelopmental disorders. *Autophagy* **17**:3908-3923 https://doi.org/10.1080/15548627.2021.1899669

17. **Courtellemont T,** Leo MGD, Gopaldass N, Mayer A (2022) CROP: a retromer- PROPPIN complex mediating membrane fission in the endo-lysosomal system. *EMBO J* **41**:e109646 https://doi.org/10.15252/embj.2021109646

18. **Cullen PJ,** Steinberg F (2018) To degrade or not to degrade: mechanisms and significance of endocytic recycling. *Nat Rev Mol Cell Biol* **19**:679-696 https://doi.org/10.1038/s41580-018-0053-7

19. **Dai J,** Li J, Bos E, Porcionatto M, Premont RT, Bourgoin S, Peters PJ, Hsu VW (2004) ACAP1 promotes endocytic recycling by recognizing recycling sorting signals. *Dev Cell* **7**:771-6 https://doi.org/10.1016/j.devcel.2004.10.002

20. **Daumke O,** Lundmark R, Vallis Y, Martens S, Butler PJG, McMahon HT (2007) Architectural and mechanistic insights into an EHD ATPase involved in membrane remodelling. *Nature* **449**:923-927 https://doi.org/10.1038/nature06173

21. **Deatherage CL,** Nikolaus J, Karatekin E, Burd CG (2020) Retromer forms low order oligomers on supported lipid bilayers. *J Biol Chem* **295**:12305-12316 https://doi.org/10.1074/jbc.ra120.013672

22. **DeLeo MG,** Berger P, Mayer A (2021) WIPI1 promotes fission of endosomal transport carriers and formation of autophagosomes through distinct mechanisms. *Autophagy* **17**:3644-3670 https://doi.org/10.1080/15548627.2021.1886830

23. **Deo R,** Kushwah MS, Kamerkar SC, Kadam NY, Dar S, Babu K, Srivastava A, Pucadyil TJ (2018) ATP-dependent membrane remodeling links EHD1 functions to endocytic recycling. *Nature communications* **9**:5187 https://doi.org/10.1038/s41467-018-07586-z

24. **Derivery E,** Sousa C, Gautier JJ, Lombard B, Loew D, Gautreau A (2009) The Arp2/3 activator WASH controls the fission of endosomes through a large multiprotein complex. *Dev Cell* **17**:712-723 https://doi.org/10.1016/j.devcel.2009.09.010

25. **Dhawan K,** Naslavsky N, Caplan S (2022) Coronin2A links actin-based endosomal processes to the EHD1 fission machinery. *Mol Biol Cell* **33**:ar107 https://doi.org/10.1091/mbc.e21-12-0624

26. **Dhawan K,** Naslavsky N, Caplan S (2020) Sorting nexin 17 (SNX17) links endosomal sorting to Eps15 homology domain protein 1 (EHD1)–mediated fission machinery. *J Biol Chem* **295**:3837-3850 https://doi.org/10.1074/jbc.ra119.011368

27. **Dooley HC,** Razi M, Polson HEJ, Girardin SE, Wilson MI, Tooze SA (2014) WIPI2 links LC3 conjugation with PI3P, autophagosome formation, and pathogen clearance by recruiting Atg12-5-16L1. *Mol Cell* **55**:238-252 https://doi.org/10.1016/j.molcel.2014.05.021

28. **Dove SK,** Piper RC, Mcewen RK, Yu JW, King MC, Hughes DC, Thuring J, Holmes AB, Cooke FT, Michell RH, *et al.* (2004) Svp1p defines a family of phosphatidylinositol 3,5-bisphosphate effectors. *EMBO J* **23**:1922-1933 https://doi.org/10.1038/sj.emboj.7600203

29. **Frisby D,** Murakonda AB, Ashraf B, Dhawan K, Almeida-Souza L, Naslavsky N, Caplan S (2024) Endosomal actin branching, fission, and receptor recycling require FCHSD2 recruitment by MICAL-L1. *Mol Biol Cell* **35**:ar144 https://doi.org/10.1091/mbc.e24-07-0324

30. **Gautier R,** Douguet D, Antonny B, Drin G (2008) HELIQUEST: a web server to screen sequences with specific alpha-helical properties. *Bioinformatics* **24**:2101-2102 https://doi.org/10.1093/bioinformatics/btn392

31. **Giridharan SSP,** Luo G, Rivero-Rios P, Steinfeld N, Tronchere H, Singla A, Burstein E, Billadeau DD, Sutton MA, Weisman LS (2022) Lipid kinases VPS34 and PIKfyve coordinate a phosphoinositide cascade to regulate retriever-mediated recycling on endosomes. *eLife* **11**:e69709 https://doi.org/10.7554/elife.69709

32. **Gokool S,** Tattersall D, Seaman MNJ (2007) EHD1 interacts with retromer to stabilize SNX1 tubules and facilitate endosome-to-Golgi retrieval. *Traffic* **8**:1873-1886 https://doi.org/10.1111/j.1600-0854.2007.00652.x

33. **Gomez TS,** Billadeau DD (2009) A FAM21-containing WASH complex regulates retromer-dependent sorting. *Dev Cell* **17**:699-711 https://doi.org/10.1016/j.devcel.2009.09.009

34. **Gopaldass N,** Chen KE, Collins B, Mayer A (2024) Assembly and fission of tubular carriers mediating protein sorting in endosomes. *Nat Rev Mol Cell Biol* **25**:765-783 https://doi.org/10.1038/s41580-024-00746-8

35. **Gopaldass N,** Fauvet B, Lashuel H, Roux A, Mayer A (2017) Membrane scission driven by the PROPPIN Atg18. *EMBO J* **36**:3274-3291 https://doi.org/10.15252/embj.201796859

36. **Gopaldass N,** Leo MGD, Courtellemont T, Mercier V, Bissig C, Roux A, Mayer A (2023) Retromer oligomerization drives SNX-BAR coat assembly and membrane constriction. *EMBO J* **42**:e112287 https://doi.org/10.15252/embj.2022112287

37. **Gopaldass N,** Mayer A (2024) PROPPINs and membrane fission in the endo- lysosomal system. *Biochem Soc Trans* **52**:1233-1241 https://doi.org/10.1042/bst20230897

38. **Grant B,** Zhang Y, Paupard MC, Lin SX, Hall DH, Hirsh D (2001) Evidence that RME- 1, a conserved C. elegans EH-domain protein, functions in endocytic recycling. *Nat Cell Biol* **3**:573-579 https://doi.org/10.1038/35078549

39. **Guan J,** Stromhaug PE, George MD, Habibzadegah-Tari P, Bevan A, Dunn WA, Klionsky DJ (2001) Cvt18/Gsa12 Is Required for Cytoplasm-to-Vacuole Transport, Pexophagy, and Autophagy in*Saccharomyces cerevisiae*and*Pichia pastoris*. *Mol Biol Cell* **12**:3821-3838 https://doi.org/10.1091/mbc.12.12.3821

40. **Guo Q,** Chen K, Gimenez-Andres M, Jellett AP, Gao Y, Simonetti B, Liu M, Danson CM, Heesom KJ, Cullen PJ, *et al.* (2024) Structural basis for coupling of the WASH subunit FAM21 with the endosomal SNX27–Retromer complex. *Proc Natl Acad Sci* **121**:e2405041121 https://doi.org/10.1073/pnas.2405041121

41. **Harbour ME,** Breusegem SY, Seaman MNJ (2012) Recruitment of the endosomal WASH complex is mediated by the extended “tail” of Fam21 binding to the retromer protein Vps35. *Biochem J* **442**:209-220 https://doi.org/10.1042/bj20111761

42. **Healy MD,** McNally KE, Butkovič R, Chilton M, Kato K, Sacharz J, McConville C, Moody ERR, Shaw S, Planelles-Herrero VJ, *et al.* (2023) Structure of the endosomal Commander complex linked to Ritscher-Schinzel syndrome. *Cell* **186**:2219-2237.e29. https://doi.org/10.1016/j.cell.2023.04.003

43. **Healy MD,** Sacharz J, McNally KE, McConville C, Tillu VA, Hall RJ, Chilton M, Cullen PJ, Mobli M, Ghai R, *et al.* (2022) Proteomic identification and structural basis for the interaction between sorting nexin SNX17 and PDLIM family proteins. *Structure* **30**:1590-1602.e6. https://doi.org/10.1016/j.str.2022.10.001

44. **Hierro A,** Rojas AL, Rojas R, Murthy N, Effantin G, Kajava AV, Steven AC, Bonifacino JS, Hurley JH (2007) Functional architecture of the retromer cargo- recognition complex. *Nature* **449**:1063-7 https://doi.org/10.1038/nature06216

45. **Hooy RM,** Iwamoto Y, Tudorica DA, Ren X, Hurley JH (2022) Self-assembly and structure of a clathrin-independent AP-1:Arf1 tubular membrane coat. *Sci Adv* **8**:eadd3914 https://doi.org/10.1126/sciadv.add3914

46. **Jeffries TR,** Dove SK, Michell RH, Parker PJ (2004) PtdIns-specific MPR Pathway Association of a Novel WD40 Repeat Protein, WIPI49. *Mol Biol Cell* **15**:2652-2663 https://doi.org/10.1091/mbc.e03-10-0732

47. **Jia D,** Gomez TS, Billadeau DD, Rosen MK (2012) Multiple repeat elements within the FAM21 tail link the WASH actin regulatory complex to the retromer. *Mol Biol Cell* **23**:2352-61 https://doi.org/10.1091/mbc.e11-12-1059

48. **Johannes L,** Wunder C, Bassereau P (2014) Bending "on the rocks"--a cocktail of biophysical modules to build endocytic pathways. *Cold Spring Harb Perspect Biol* **6**:a016741-a016741 https://doi.org/10.1101/cshperspect.a016741

49. **Kendall AK,** Chandra M, Xie B, Wan W, Jackson LP (2022) Improved mammalian retromer cryo-EM structures reveal a new assembly interface. *J Biol Chem* **298**:e102523 https://doi.org/10.1016/j.jbc.2022.102523

50. **Kendall AK,** Xie B, Xu P, Wang J, Burcham R, Frazier MN, Binshtein E, Wei H, Graham TR, Nakagawa T, *et al.* (2020) Mammalian Retromer Is an Adaptable Scaffold for Cargo Sorting from Endosomes. *Structure* **28**:393-405.e4. https://doi.org/10.1016/j.str.2020.01.009

51. **Kovtun O,** Leneva N, Bykov YS, Ariotti N, Teasdale RD, Schaffer M, Engel BD, Owen DJ, Briggs JAG, Collins BM (2018) Structure of the membrane-assembled retromer coat determined by cryo-electron tomography. *Nature* **561**:561-564 https://doi.org/10.1038/s41586-018-0526-z

52. **Krick R,** Busse RA, Scacioc A, Stephan M, Janshoff A, Thumm M, Kühnel K (2012) Structural and functional characterization of the two phosphoinositide binding sites of PROPPINs, a β-propeller protein family. *Proc Natl Acad Sci USA* **109**:E2042-9 https://doi.org/10.1073/pnas.1205128109

53. **Kvainickas A,** Jimenez-Orgaz A, Nägele H, Hu Z, Dengjel J, Steinberg F (2017) Cargo-selective SNX-BAR proteins mediate retromer trimer independent retrograde transport. *J Cell Biology* **216**:3677-3693 https://doi.org/10.1083/jcb.201702137

54. **Laulumaa S,** Kumpula E-P, Huiskonen JT, Varjosalo M (2024) Structure and interactions of the endogenous human Commander complex. *Nat Struct Mol Biol* **31**:925-938 https://doi.org/10.1038/s41594-024-01246-1

55. **Leneva N,** Kovtun O, Morado DR, Briggs JAG, Owen DJ (2021) Architecture and mechanism of metazoan retromer:SNX3 tubular coat assembly. *Sci Adv* **7**:eabf8598 https://doi.org/10.1126/sciadv.abf8598

56. **Li J,** Peters PJ, Bai M, Dai J, Bos E, Kirchhausen T, Kandror KV, Hsu VW (2007) An ACAP1-containing clathrin coat complex for endocytic recycling. *J Cell Biol* **178**:453-64 https://doi.org/10.1083/jcb.200608033

57. **Liang R,** Ren J, Zhang Y, Feng W (2019) Structural Conservation of the Two Phosphoinositide-Binding Sites in WIPI Proteins. *J Mol Biol* **431**:1494-1505 https://doi.org/10.1016/j.jmb.2019.02.019

58. **Lopez-Robles C,** Scaramuzza S, Astorga-Simon EN, Ishida M, Williamson CD, Banos-Mateos S, Gil-Carton D, Romero-Durana M, Vidaurrazaga A, Fernandez- Recio J, *et al.* (2023) Architecture of the ESCPE-1 membrane coat. *Nat Struct Mol Biol* **30**:958-969 https://doi.org/10.1038/s41594-023-01014-7

59. **Lucas M,** Gershlick DC, Vidaurrazaga A, Rojas AL, Bonifacino JS, Hierro A (2016) Structural Mechanism for Cargo Recognition by the Retromer Complex. *Cell* **167**:1623-1635 https://doi.org/10.1016/j.cell.2016.10.056

60. **Mallam AL,** Marcotte EM (2017) Systems-wide Studies Uncover Commander, a Multiprotein Complex Essential to Human Development. *Cell Syst* **4**:483-494 https://doi.org/10.1016/j.cels.2017.04.006

61. **Manders EMM,** Verbeek FJ, Aten JA (2011) Measurement of co-localization of objects in dual-colour confocal images. *J Microsc* **169**:375-382 https://doi.org/10.1111/j.1365-2818.1993.tb03313.x

62. **Marquardt L,** Taylor M, Kramer F, Schmitt K, Braus GH, Valerius O, Thumm M (2023) Vacuole fragmentation depends on a novel Atg18-containing retromer- complex. *Autophagy* **19**:278-295 https://doi.org/10.1080/15548627.2022.2072656

63. **Martín-González A,** Méndez-Guzmán I, Zabala-Zearreta M, Quintanilla A, García- López A, Martínez-Lombardía E, Albesa-Jové D, Acosta JC, Lucas M (2024) Selective cargo and membrane recognition by SNX17 regulates its interaction with Retriever. *EMBO Rep* **26**:470-493 https://doi.org/10.1038/s44319-024-00340-1

64. **McGough IJ,** Steinberg F, Gallon M, Yatsu A, Ohbayashi N, Heesom KJ, Fukuda M, Cullen PJ (2014) Identification of molecular heterogeneity in SNX27-retromer- mediated endosome-to-plasma-membrane recycling. *J Cell Sci* **127**:4940-4953 https://doi.org/10.1242/jcs.156299

65. **McNally KE,** Cullen PJ (2018) Endosomal Retrieval of Cargo: Retromer Is Not Alone. *Trends Cell Biol* https://doi.org/10.1016/j.tcb.2018.06.005

66. **McNally KE,** Faulkner R, Steinberg F, Gallon M, Ghai R, Pim D, Langton P, Pearson N, Danson CM, Nagele H, *et al.* (2017) Retriever is a multiprotein complex for retromer-independent endosomal cargo recycling. *Nat Cell Biol* **19**:1214-1225 https://doi.org/10.1038/ncb3610

67. **Mellman I,** Yarden Y (2013) Endocytosis and Cancer. *Cold Spring Harb Perspect Biol* **5**:a016949 https://doi.org/10.1101/cshperspect.a016949

68. **Naslavsky N,** Caplan S (2023) Advances and challenges in understanding endosomal sorting and fission. *Febs J* **290**:4187-4195 https://doi.org/10.1111/febs.16687

69. **Obara K,** Sekito T, Niimi K, Ohsumi Y (2008) The Atg18-Atg2 complex is recruited to autophagic membranes via phosphatidylinositol 3-phosphate and exerts an essential function. *Journal of Biological Chemistry* **283**:23972-23980 https://doi.org/10.1074/jbc.m803180200

70. **Pang X,** Fan J, Zhang Y, Zhang K, Gao B, Ma J, Li J, Deng Y, Zhou Q, Egelman EH, *et al.* (2014) A PH domain in ACAP1 possesses key features of the BAR domain in promoting membrane curvature. *Dev Cell* **31**:73-86 https://doi.org/10.1016/j.devcel.2014.08.020

71. **Phillips-Krawczak CA,** Singla A, Starokadomskyy P, Deng Z, Osborne DG, Li H, Dick CJ, Gomez TS, Koenecke M, Zhang J-S, *et al.* (2015) COMMD1 is linked to the WASH complex and regulates endosomal trafficking of the copper transporter ATP7A. *Mol Biol Cell* **26**:91-103 https://doi.org/10.1091/mbc.e14-06-1073

72. **Polson HEJ,** Lartigue JD, Rigden DJ, Reedijk M, Urbé S, Clague MJ, Tooze SA (2010) Mammalian Atg18 (WIPI2) localizes to omegasome-anchored phagophores and positively regulates LC3 lipidation. *Autophagy* **6**:506-522 https://doi.org/10.4161/auto.6.4.11863

73. **Proikas-Cezanne T,** Takacs Z, Dönnes P, Kohlbacher O (2015) WIPI proteins: essential PtdIns3*P* effectors at the nascent autophagosome. *J Cell Sci* **128**:207-217 https://doi.org/10.1242/jcs.146258

74. **Proikas-Cezanne T,** Waddell S, Gaugel A, Frickey T, Lupas A, Nordheim A. (2004) WIPI-1α (WIPI49), a member of the novel 7-bladed WIPI protein family, is aberrantly expressed in human cancer and is linked to starvation-induced autophagy. *Oncogene* **23**:9314-9325 https://doi.org/10.1038/sj.onc.1208331

75. **Puri C,** Vicinanza M, Ashkenazi A, Gratian MJ, Zhang Q, Bento CF, Renna M, Menzies FM, Rubinsztein DC (2018) The RAB11A-Positive Compartment Is a Primary Platform for Autophagosome Assembly Mediated by WIPI2 Recognition of PI3P-RAB11A. *Dev Cell* **45**:114-131.e8. https://doi.org/10.1016/j.devcel.2018.03.008

76. **Purushothaman LK,** Arlt H, Kuhlee A, Raunser S, Ungermann C (2017) Retromer- driven membrane tubulation separates endosomal recycling from Rab7/Ypt7- dependent fusion. *Mol Biol Cell* **28**:783-791 https://doi.org/10.1091/mbc.e16-08-0582

77. **Ren X,** Farias GG, Canagarajah BJ, Bonifacino JS, Hurley JH (2013) Structural basis for recruitment and activation of the AP-1 clathrin adaptor complex by Arf1. *Cell* **152**:755-67 https://doi.org/10.1016/j.cell.2012.12.042

78. **Rivero-Ríos P,** Weisman LS (2022) Roles of PIKfyve in multiple cellular pathways. *Curr Opin Cell Biol* **76**:102086 https://doi.org/10.1016/j.ceb.2022.102086

79. **Sachse M,** Urbe S, Oorschot V, Strous GJ, Klumperman J (2002) Bilayered clathrin coats on endosomal vacuoles are involved in protein sorting toward lysosomes. *Mol Biol Cell* **13**:1313-28 https://doi.org/10.1091/mbc.01-10-0525

80. **Schreij AMA,** Fon EA, McPherson PS (2016) Endocytic membrane trafficking and neurodegenerative disease. *Cell Mol Life Sci* **73**:1529-1545 https://doi.org/10.1007/s00018-015-2105-x

81. **Seaman MN,** Harbour ME, Tattersall D, Read E, Bright N (2009) Membrane recruitment of the cargo-selective retromer subcomplex is catalysed by the small GTPase Rab7 and inhibited by the Rab-GAP TBC1D5. *J Cell Sci* **122**:2371-82 https://doi.org/10.1242/jcs.048686

82. **Seaman MNJ** (2021) The Retromer Complex: From Genesis to Revelations. *Trends Biochem Sci* **6**:9075 https://doi.org/10.1016/j.tibs.2020.12.009

83. **Shortill SP,** Frier MS, Wongsangaroonsri P, Davey M, Conibear E (2022) The VINE complex is an endosomal VPS9-domain GEF and SNX-BAR coat. *eLife* **11**:e77035 https://doi.org/10.7554/elife.77035

84. **Simonetti B,** Danson CM, Heesom KJ, Cullen PJ (2017) Sequence-dependent cargo recognition by SNX-BARs mediates retromer-independent transport of CI-MPR. *J Cell Biol* **216**:3695-3712 https://doi.org/10.1083/jcb.201703015

85. **Simonetti B,** Guo Q, Gimenez-Andres M, Chen KE, Moody ERR, Evans AJ, Chandra M, Danson CM, Williams TA, Collins BM, *et al.* (2022) SNX27-Retromer directly binds ESCPE-1 to transfer cargo proteins during endosomal recycling. *PLoS Biol* **20**:e3001601 https://doi.org/10.1371/journal.pbio.3001601

86. **Simonetti B,** Paul B, Chaudhari K, Weeratunga S, Steinberg F, Gorla M, Heesom KJ, Bashaw GJ, Collins BM, Cullen PJ (2019) Molecular identification of a BAR domain-containing coat complex for endosomal recycling of transmembrane proteins. *Nat Cell Biol* **21**:1219-1233 https://doi.org/10.1038/s41556-019-0393-3

87. **Simunovic M,** Bassereau P, Voth GA (2018) Organizing membrane-curving proteins: the emerging dynamical picture. *Curr Opin Struct Biol* **51**:99-105 https://doi.org/10.1016/j.sbi.2018.03.018

88. **Simunovic M,** Manneville J-B, Renard H-F, Evergren E, Raghunathan K, Bhatia D, Kenworthy AK, Voth GA, Prost J, McMahon HT, *et al.* (2017) Friction Mediates Scission of Tubular Membranes Scaffolded by BAR Proteins. *Cell* **170**:172-184.e11. https://doi.org/10.1016/j.cell.2017.05.047

89. **Singla A,** Boesch DJ, Fung HYJ, Ngoka C, Enriquez AS, Song R, Kramer DA, Han Y, Banarer E, Lemoff A, *et al.* (2024) Structural basis for Retriever-SNX17 assembly and endosomal sorting. *Nat Commun* **15**:10193 https://doi.org/10.1038/s41467-024-54583-6

90. **Solinger JA,** Rashid H-O, Prescianotto-Baschong C, Spang A (2020) FERARI is required for Rab11-dependent endocytic recycling. *Nat Cell Biol* **59**:237-12 https://doi.org/10.1038/s41556-019-0456-5

91. **Steinberg F,** Gallon M, Winfield M, Thomas EC, Bell AJ, Heesom KJ, Tavaré JM, Cullen PJ (2013) A global analysis of SNX27–retromer assembly and cargo specificity reveals a function in glucose and metal ion transport. *Nat Cell Biol* **15**:461-471 https://doi.org/10.1038/ncb2721

92. **Steinberg F,** Heesom KJ, Bass MD, Cullen PJ (2012) SNX17 protects integrins from degradation by sorting between lysosomal and recycling pathways. *J Cell Biol* **197**:219-230 https://doi.org/10.1083/jcb.201111121

93. **Temkin P,** Lauffer B, Jäger S, Cimermancic P, Krogan NJ, von Zastrow M (2011) SNX27 mediates retromer tubule entry and endosome-to-plasma membrane trafficking of signalling receptors. *Nat Cell Biol* **13**:715-721 https://doi.org/10.1038/ncb2252

94. **Vicinanza M,** Campli AD, Polishchuk E, Santoro M, Tullio GD, Godi A, Levtchenko E, Leo MGD, Polishchuk R, Sandoval L, *et al.* (2011) OCRL controls trafficking through early endosomes via PtdIns4,5P_2-dependent regulation of endosomal actin. *EMBO J* **30**:4970-4985

https://doi.org/10.1038/emboj.2011.354

95. **Vicinanza M,** Korolchuk IV, Ashkenazi A, Puri C, Menzies FM, Clarke JH, Rubinsztein DC (2015) PI(5)P Regulates Autophagosome Biogenesis. *Mol Cell* **57**:219-234 https://doi.org/10.1016/j.molcel.2014.12.007
96. **Watanabe Y,** Kobayashi T, Yamamoto H, Hoshida H, Akada R, Inagaki F, Ohsumi Y, Noda NN (2012) Structure-based analyses reveal distinct binding sites for Atg2 and phosphoinositides in Atg18. *J Biol Chem* **287**:31681-31690 https://doi.org/10.1074/jbc.m112.397570
97. **van Weering JRT,** Sessions RB, Traer CJ, Kloer DP, Bhatia VK, Stamou D, Carlsson SR, Hurley JH, Cullen PJ (2012) Molecular basis for SNX-BAR-mediated assembly of distinct endosomal sorting tubules. *EMBO J* **31**:4466-4480 https://doi.org/10.1038/emboj.2012.283
98. **Zhang Y,** Pang X, Li J, Xu J, Hsu VW, Sun F (2021) Structural insights into membrane remodeling by SNX1. *Proc Natl Acad Sci* **118**:e2022614118 https://doi.org/10.1073/pnas.2022614118
99. **Zieger M,** Mayer A (2012) Yeast vacuoles fragment in an asymmetrical two-phase process with distinct protein requirements. *Mol Biol Cell* **23**:3438-3449 https://doi.org/10.1091/mbc.e12-05-0347

Peer reviews

Reviewer #1 (Public review):

WIPI1 is a PROPPIN family protein that has been implicated in Retromer-mediated membrane fission events. Although the cargos that it has been tested to be important for are diverse, one of the cargos that is unaffected is Beta1-Integrin. This leads the authors to assess another PROPPIN family protein - WIPI2, which is a homolog of WIPI1. KD using siRNA is effective and had no consequences on LAMP1, EGFR trafficking or GLUT1 trafficking. Integrin-B1, however, had a large and significant defect in its recycling from the endosome, with a clear endosomal colocalisation. Complementation experiments with WT WIPI2 recovered the phenotype, but various mutant WIPI2 complements resulted in elongated tubules, and there was also a dominant negative effect of the mutant. Integrin is a classic retreiver cargo, so the authors rationalise that WIPI2 may be playing a role with retreiver that WIPI1 plays with retromer. To assess this, they perform a set of immunoprecipitations. SNX17, the retreiver-associated sorting nexin, co-IPs with WIPI2 in a VPS26C-dependent manner. VPS26C but not VPS26 co-IPs with WIPI2, and the reciprocal with WIPI1. These interactions were not present for the FSSS mutation of WIPI2. WIPI2 localises to Rab11 endosomes mainly, as does retriever. Mutations of WIPI2 not only affected WIPI2 localisation, but also VPS35L mutations, indicating that there is a functional relationship between the two.

On the whole, I find the manuscript compelling. The manuscript is very clearly written, the results are convincing and well performed. The flow of experiments is logical, and although not comprehensive in the subsequent mechanistic understanding, the fundamental findings are important and convincing. My comments below are, on the whole, minor and are intended to support the communication of the findings to the field.

(1) The IP interaction data were convincing; however, for me and some others, an interaction is only convincing when performed in vitro, and understood at a structural level. I do not suggest the authors do that in this case; however, I think, at a minimum, some sensible moderation of claims would be useful here.

(2) I found the final localisation data and its interpretation confusing. My interpretation of that data would not be that the retreiver is relocalised, but rather that there is less of both recruited to the membrane and the remaining localisation distribution is shifted. In addition, I am not quite sure of the model here - is the idea that WIPI2 recruits retreiver, if that is the case, I find it hard to resolve with its role as a mediator of fission. Clarity would be appreciated here.

(3) I am concerned that the repeats being compared for statistical analysis are not biological repeats but technical repeats (cells in the same experiment). I should think the idea of the statistical comparison is to show experimental reproducibility and variability across biological repeats. Therefore, I would expect an appropriate number of biological repeats (3 or more minimum), to be the data compared in the statistical analysis and graphs. I think it is appropriate to average the technical repeats from each biological repeat. I find these to be useful resources *https://doi.org/10.1083/jcb.202401074*, *https://doi.org/10.1083/jcb.200611141*

https://doi.org/10.7554/eLife.109403.1.sa2

Reviewer #2 (Public review):

Summary:

The manuscript from De Leo and Mayer presents evidence that the PROPPIN protein, WIPI2, associates with the Retriever complex, and is required for the proper transport of the SNX17-Retriever cargo, beta1-integrin. This finding fits with prior papers from the Mayer lab, which showed that a related PROPPIN, WIPI1, is required for the transport of some SNX27-Retromer cargo, including GLUT1. The retromer and retriever complexes are architecturally similar. Importantly, they act at the same endosomes, and each transports cargo from endosomes to the plasma membrane. Thus, the possibility that each also requires a structurally related PROPPIN is of interest. However, the manuscript is incomplete, and the main claims are only partially supported.

Strengths:

The topic that PROPPIN proteins are important for the function of the Retromer and Retriever complexes expands our view of the trafficking complex.

Weaknesses:

Many important controls are missing. Several points that are made in the manuscript are only supported through a single approach.

https://doi.org/10.7554/eLife.109403.1.sa1

Reviewer #3 (Public review):

Summary:

The manuscript of Mayer and colleagues analyzes the function of WIPI proteins in mammalian cells. The authors previously identified CROP as a complex consisting of WIPI1 and the retromer complex, primarily in yeast cells. In mammalian cells, both WIPI1 and WIPI2 exist, whereas retromer has a homologous complex termed retriever. They now find that WIPI2 can form a complex with retriever subunits. They named this complex CROP2. Their data further indicate that CROP2 and CROP1 have distinct substrate specificities as knockdown of CROP2 subunits affects beta1 integrin sorting, whereas knockdown of CROP1 affects EGFR and GLUT1. They further identify a similar sequence (FSSS) in both WIPI1 and WIPI2, which is required for their specific binding to retromer and retriever.

Strengths:

CROP1 and CROP2 seem to use similar features for their formation, and have different substrates, which is convincingly shown.

Weaknesses:

The analysis lacks information that this is a complex as claimed. It can be deduced from the interaction analysis, but was not shown.

https://doi.org/10.7554/eLife.109403.1.sa0